The Manager as a Nexus of Relational Contracts

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Abstract

Why and when do managers replace markets in allocating resources? We develop a general equilibrium model of an exchange economy in which centralized monitors partly supplant anonymous matching markets in coordinating assignments between buyers and producers. In our model, managers emerge as nexuses of relational contracts to mitigate a fundamental conflict between incentive provision and flexible allocations in fluctuating environments. We find that managerial coordination strengthens relational incentives and reduces misallocation, but leads to double marginalization of incentive rents. Because of this trade-off, market participants sort into decentralized market exchange and managerial coordination. Their optimal choice depends on factors such as demand volatility, specialization needs, market size, market tightness, and market-level reputation effects. Our model provides novel insights into the equilibrium structure, boundaries, and economic impacts of firm-like organizations, explaining when and why managerial hierarchies emerge from market-based mechanisms.

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1 Introduction

The presence of *managers* (i.e. centralized monitor-coordinators) has long been considered a key feature of firm-like economic organization. In a seminal article, Alchian and Demsetz (1972) proposed that the essence of the firm is the presence of a central contracting party that monitors performance and excludes underperforming team members from future participation. Jensen and Meckling (1976, 310–311) expanded on this view and suggested that "contractual relations are the essence of the firm, not only with employees but with suppliers, customers, creditors, and so on." Demsetz (1988, 154) concurred that "the firm properly viewed is a 'nexus' of contracts" and elaborated that the key attributes of firm-like organization are continuity of association, specialization, and managerial direction. Yet, microeconomic foundations for these influential ideas have been illusive.

In this paper, we develop a simple general equilibrium model of an exchange economy in which monitors endogenously emerge as centralized resource allocators. Our theory builds on a growing literature that views *relational* contracts—collaborations sustained by repeated interactions—as ubiquitous within and between organizations (Baker, Gibbons and Murphy 2002; Gibbons and Henderson 2012; Macchiavello and Morjaria 2023; Muehlheusser, Fahn and MacLeod 2023). We show that managers emerge as nexuses of *relational* contracts.

Why does the *visible hand* of managers supplant the *invisible hand* of markets in allocating resources? In our theory, they do so to redress a fundamental conflict between incentive provision and flexible assignments in fluctuating environments. We will show why and when this tension causes managers to replace anonymous matching markets in coordinating allocations. We will then characterize how managers affect market efficiency, the division of labor, matching market flows, and the distribution of economic rents. Our results can be used to understand a variety of real-world organizations, including professional service firms that monitor workers and flexibly assign them to clients—such as those offering security, law, accounting, HR, and IT services (Abraham and Taylor 1996)—and global sourcing companies like Li & Fung that fulfill retailer demand by coordinating and monitoring suppliers (Belavina and Girotra 2012).

The basic structure of our model is similar to Shapiro and Stiglitz (1984). In the model, producers cannot be compelled to exert effort using written legal contracts, so buyers must motivate producers using future surplus in relational contracts. Buyers and producers meet and enter contracts in frictionless matching markets without search delays or hidden payoff-relevant information, as in MacLeod and Malcomson (1989) and Board and Meyer-ter Vehn (2015). Departing from prior models, we introduce a new feature—fluctuating buyer demand—to capture a realistic and relevant aspect of repeated business interactions (Macchiavello and Morjaria 2015). We also introduce managers who cannot themselves produce, but can monitor a large number of producers. In equilibrium, managers enter multiple relational contracts on both sides of the market, aggregate fluctuating demand, coordinate assignments, and keep track of producer performance.

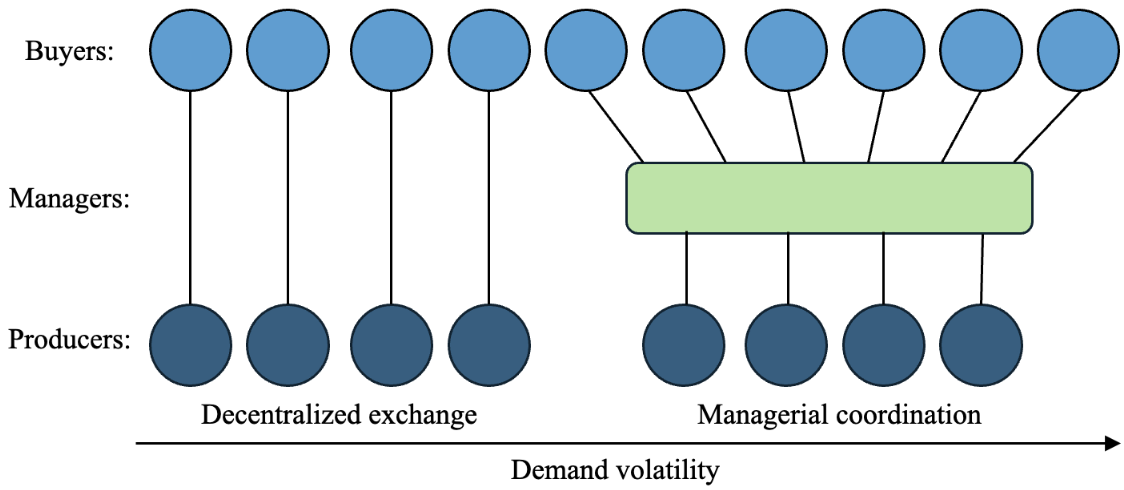
We show that the use of managers involves a trade-off between demand aggregation and double marginalization. The benefit of involving managers is that they aggregate fluctuating buyer demand and thereby prevent producers from becoming idle or unmatched. Therefore, managers lower the incentive rent needed to motivate producers to exert effort. However, managers must be motivated by an additional incentive rent. The cost of managers is thus a form of double marginalization.[1]

Because of this trade-off, buyers sort into managerial coordination and decentralized exchange. Their optimal choice depends on individual characteristics, such as their individual demand volatility. It also depends on market-level characteristics, such as market tightness and market size. If the matching market is sufficiently tight, managers emerge as contracting nexuses. Buyers whose business needs are short-lived contract with managers who coordinate assignments. The remaining buyers contract directly with producers in a sea of decentralized pairings. Figure 1 illustrates the relational contracting networks that emerge in the unique steady-state equilibrium.

We derive a rich set of comparative statics regarding the steady-state determinants and impacts

[1]In the classic literature in industrial organization, double marginalization occurs when both upstream and downstream parties possess market power and use restrictive linear contracts. This form of double marginalization can be eliminated by vertical integration of pricing decision rights, resale-price maintenance, or non-linear pricing (Tirole 1988). In our model, double marginalization instead arises from the non-verifiability of contract performance and the non-transferability of production and monitoring capabilities, as in models of middleman margins (Biglaiser and Friedman 1994; Bardhan, Mookherjee and Tsumagari 2013). Conventional remedies do not eliminate this form of double marginalization.

Figure 1: Relational contracting networks in steady-state equilibrium



of managerial coordination. First, when compared to managed producers, directly contracted producers have higher average incentive pay, more dispersed pay, higher separation rates, and higher idleness. Second, the introduction of managers into an economy benefits buyers with fluctuating demand, but reduces the pay of producers initially earning pay premiums. As such, managerial coordination is both efficiency-enhancing and redistributive. Third, managerial coordination is profitable only if there is a sufficiently large market of buyers and producers. In other words, managerial coordination requires economies of scale. We also characterize potential market-level spillover effects from managerial coordination.

In two extensions, we analyze the relationship between managerial coordination, specialization, and the strength of market-level reputation effects. We show that managed producers are more specialized and that managerial coordination increases producer specialization. Moreover, buyers are more likely to choose managerial coordination if there are gains from producer specialization and if the managers have reputation concerns arising from word-of-mouth communication about manager performance. We also show that, unlike intermediation that mitigates search frictions or adverse selection, managerial coordination may *increase* as communication frictions fall. The reason is that ease of communication may allow buyers to better coordinate on collective punishments against managers who shirk. This lowers manager incentive rents, so managerial coordination becomes more attractive.

Throughout the paper, we discuss connections to the empirical literature on professional service firms, which has recently blossomed (e.g, Garicano and Hubbard 2009; Goldschmidt and Schmieder 2017; Felix and Wong 2024). As we shall show, our model generates realistic predictions for the firm-level and market-level drivers of professional service outsourcing, as well as for the effects of professional service firms on both workers and the labor market as a whole. Although many theories exist to explain the labor boundaries of firms, our model provides a novel and useful lens to understand their equilibrium structure, boundaries, and impacts. We are the first, for example, to show how professional service firms may both reduce worker rents and improve efficiency purely because of their role as centralized monitor-coordinators.

1.1 Related Theory

The game-theoretic foundation for our theory comes from a growing literature on relational contracts in matching markets (Shapiro and Stiglitz 1984; MacLeod and Malcomson 1989, 1998; Yang 2008; Board and Meyer-ter Vehn 2015; Fahn 2017; Powell 2019; Fahn and Murooka 2022; Li 2022).[2] In particular, our modeling approach builds directly on MacLeod and Malcomson (1989) and Board and Meyer-ter Vehn (2015). The novelty here is to introduce fluctuating demand and *managers* into frictionless markets for relational contracts. We show that by aggregating fluctuating demand and coordinating assignments, managers can ease the tension between incentive provision and flexible allocations, and thereby enable more trade and specialization. In related work, Board (2011) shows that the tension between relational incentives and flexible allocations can lead buyers to restrict their number of trading partners. Andrews and Barron (2016) show that this tension can cause dynamic allocations to depend on payoff-irrelevant past performance. Li and Powell (2020) highlight that a similar tension can lead agents to interact in multiple activities. To our knowledge, we are the first to show that this tension gives rise to centralized monitor-coordinators in exchange economies.[3]

[2]For surveys of recent work on relational contracts in economics, see MacLeod (2007), Malcomson (2013), Gil and Zanarone (2017), and Macchiavello and Morjaria (2023). The study of relational contracts also has a multidisciplinary pedigree: in anthropology, see Geertz (1962, 1978); in sociology, see Macaulay (1963) and Granovetter (1985); in law, see Macneil (1978); in strategy, see Dyer and Singh (1998).

[3]Recent models of relational contracting intermediaries emphasize very different mechanisms. Fong and Li (2017) show that relational contracts can be deepened by the presence of a non-strategic supervisor carrying out

Our work complements the two main strands of formal work on the theory of the firm. First, a large set of theories study the optimal allocation of asset ownership using variants of the property rights approach proposed by Grossman and Hart (1986).[4] We view our work as complementary, since it can be combined with the property rights approach to study the joint ownership and contracting structure of firms. For example, Baker, Gibbons and Murphy (2002) provide models in which asset ownership alters the strength of relational contracts. Second, another large group of theories explore the nature of employment contracts. For example, some study the implications of contract-writing and bargaining difficulties (e.g. Wernerfelt 1997, 2015, 2016; Hart and Moore 2008; Levin and Tadelis 2010; Tadelis and Williamson 2012). Others study the differences between relational and spot contracts (e.g. Simon 1951; Bolton and Rajan 2003; Raith 2022). Our work is also complementary to this strand of the literature, since we instead characterize how *networks* of relational contracts emerge from random matching between agents in exchange economies. We view the emergence of centralized contracting agents as an important but as yet understudied aspect of firm-like organization.

Ideas related to ours can be found in the literatures on supply chain management and market institutions. In the supply chain management literature, Belavina and Girotra (2012) develop a model of relational intermediaries with two buyers, two sellers, and non-transferable utilities. Our transferable utility model is similar but enables a fuller characterization of the determinants and welfare consequences of intermediation. In the literature on market institutions, Milgrom, North and Weingast (1990) use a model of repeated Prisoner's Dilemma games to show that private judges in medieval trade can create trust with less observability than reputation systems. Our paper follows a similar logic, but our focus is not the historical role of legal and regulatory institutions. We instead view the problem of trust as pervasive despite the development of modern institutions and suggest that it leads to managerial coordination in many sectors of the

subjective performance reviews. Troya-Martinez and Wren-Lewis (2023) explore a model where a manager may receive kick-backs and show that such managers can improve relational contracts in environments with commitment difficulty. Biglaiser and Friedman (1994) and Bardhan, Mookherjee and Tsumagari (2013) study models in which middlemen with reputations have a larger incentive to monitor and are in a better position to learn about quality than an ordinary buyer does because they buy a larger proportion of the producers' goods. Our model instead focuses on how managers alleviate the tension between flexible allocation and incentive provision.

[4]See Gibbons (2005) for a comprehensive treatment of related formal theories.

contemporary economy.[5]

The rest of the paper proceeds as follows. Section 2 introduces our model. Section 3 characterizes and compares direct and managed contracts, taking matching market conditions as given. Section 4 derives the steady-state equilibrium by endogenizing matching market flows. Section 5 pursues extensions. Section 6 concludes.

2 Model

Basics. Time is discrete and infinite, $t \in \{0, 1, \ldots\}$. There are a unit mass of buyers indexed by i who may demand services. There is a measure n of producers indexed by j who provide services. The measure of producers is determined by endogenous entry at the start of each period subject to entry cost $C > 0$, which represents the producers' training or opportunity cost. There is a finite number K of managers indexed by k who neither directly demand nor provide services. All players are infinitely-lived and have a common discount factor $\delta \in (0, 1)$.

Demand realization. After the endogenous entry of producers, the service demand of each buyer i, denoted $d_{it} \in \{0, 1\}$, is realized and publicly observed.[6] Each buyer's demand d_{it} is independently drawn at the beginning of each period t following a Markov process. The demand-switching probabilities are each buyer's publicly known type $\alpha_i = (\alpha_{1i}, \alpha_{0i})$: With probability α_{1i}, buyer i's demand switches from 1 to 0. With probability α_{0i}, buyer i's demand switches back from 0 to 1. The distribution of buyers types is a distribution F on $[0, 1] \times [0, 1]$.

Matching. Both buyers and managers can offer contracts to producers in a producer market. In addition, buyers can offer contracts to managers in a manager market. Each contract is a contingency plan specifying compensation, effort levels, and the probability of continuation

[5]Henderson and Churi (2019) offer a related perspective and traces the history of innovation and trust from medieval guides, early corporations, New Deal administrative agencies, and Internet-age companies such as Uber. For related papers on market institutions, see Greif (1993, 2006), Greif, Milgrom and Weingast (1994), Calvert (1995), Ramey and Watson (2002), Fafchamps (2004), and Budish (2023).

[6]To focus on the relational incentives, we abstract from any adverse selection problem in the model. Therefore, when contracting with buyers, producers know whom they are dealing with and have the correct expectation of how the relationships will go.

when demand switches. Following Board and Meyer-ter Vehn (2015), matching is frictionless, meaning that all offers are accepted subject to participation constraints.

In the producer market, matching is random and anonymous, meaning that all unmatched producers have the same probability of being matched with a given buyer or manager regardless of their histories. We rule out on-the-job search by assuming that matched producers do not receive contract offers.[7]

In the manager market, matching is not anonymous, so matching probabilities may depend on the observable history of a manager. Since managers cannot produce by themselves, they fulfill buyer contract requirements by entering contracts with producers.

We assume that a manager can match with a continuum of buyers and producers. In contrast, in any period, buyers can match with at most one producer or manager, and producers can match with at most one buyer or manager.[8] Matching in the manager market precedes matching in the producer market, so managers can always fulfill the service demands of their matched buyers if there is an excess of producers.

Direct contracting. If a buyer matches with a producer, the remainder of the period proceeds as follows. First, the buyer makes a payment $w_t \geq 0$ to the producer.[9] The producer then chooses an effort level, denoted by $e_t \in \{0, 1\}$. The cost of effort is given by $c(e_t)$, where $c(0) = 0$ and $c(1) = c > 0$. The effort generates an output for the buyer only if demand is positive, so $y_{it} = y d_{it} e_t$. Effort and output are observable by the buyer but are not verifiable by a court. We assume that $y > \frac{c}{\delta} + (1 - \delta)C$, so that for buyers with unchanging and positive demand, there is always enough surplus to incentivize producer effort.[10]

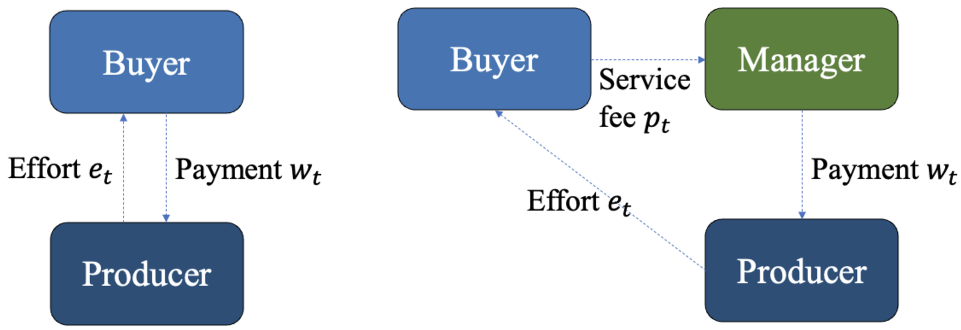
[7]Board and Meyer-ter Vehn (2015) analyzes relational contracts in a frictionless matching market where matched producers may receive on-the-job offers. They show that this leads to heterogeneous productivity across otherwise identical matches. We rule out this possibility for simplicity.

[8]This assumption rules out the possibility that producers and buyers become managers themselves.

[9]The assumption that pay is always non-negative is realistic in many real-world settings. It also significantly simplifies the analysis. Appendix B proves that imposing this assumption in our model is without loss. Due to this limited liability assumption, equilibrium pay is always zero during periods when demand is zero (see Lemma A.1 for details), so one can compare the value of different stationary contracts by comparing pay when demand is one.

[10]Here we do not allow for ex-post bonus payments, which play in important role in MacLeod and Malcomson (1998) and Levin (2003). This assumption is not important: There is an excess of producers, so buyers in our model have all the bargaining power. Therefore, for any contract with bonuses, there is a weakly more profitable contract for buyers without bonuses. Intuitively, the buyer prefers to pay the producer at the latest possible moment before e_t.

Figure 2: Structure of direct and managed contracts



(a) Direct contract

(b) Managed contract

Managed contracting. If a buyer matches with a manager, she pays a service fee $p_t \geq 0$ to the manager. The manager chooses to assign one or none of its producers to the buyer. The manager pays w_t to the producer. The producer then exerts costly effort e_t and produces output y_{it} for the buyer.[11] Effort and output are observable by both the buyer and the manager, but are not verifiable by a court.

Separation. Either party in a match can choose to terminate their contract and separate from each other both after demand realization and after production. If separation occurs after demand realization, both parties can participate in the producer or manager market matching in the current period. However, if separation occurs after production, they need to wait till the next period to rematch.

Payoffs. In a direct contract, the buyer's payoff is $y_{it} - w_t$ in each period t. In an managed contract, the buyer's payoff is $y_{it} - p_t$. The manager's payoff per service demand is $p_t - w_t$. The producer's payoff is $w_t - c(e_t)$.

It is therefore weakly profitable to shift the bonus b_{t-1} into the next period's expected pay, such that it is paid only when the next period's demand is one and the producer is retained. Since effort is always perfectly observable in our setting, the proof for a similar result in Board and Meyer-ter Vehn (2015) can be easily extended to our setting despite the presence of limited liability. See Appendix B for a formal proof.

[11] Similarly, we do not allow for ex-post service fees. This is unimportant for the same reason that ex-post bonuses are assumed away; buyers can immediately rematch with new managers, and contracts with ex-post service fees are weakly less profitable for buyers.

Remark. This model has two key features that are not present in standard models of relational contracting in frictionless matching markets, such as Shapiro and Stiglitz (1984). First, we allow buyers to have fluctuating demand. Second, we introduce managers who can enter relational contracts with a multitude of agents on both sides of the market. These managers do not have intrinsic demand or productive ability. What they can do, however, is to match with a large set of buyers and producers and monitor the performance of all of their matched producers. As we shall show, when demand is volatile and the cost of idleness is high, managers can sustain cheaper relational contracts on both sides of the market by reassigning their producers across buyers. Buyers may therefore prefer managerial coordination to direct contracting.

Example (Professional service firms)**.** Consider an entrepreneur (*buyer*) who needs a service performed. She can fulfill the demand either by employing an in-house worker (*producer*) or contracting it out to a professional service firm (*manager*) that allocates its employees to clients. Entrepreneurs may make these employ-or-outsource decisions for various professional services, including cleaning, security, accounting, legal, IT, and HR services.[12] These decisions determine the labor boundary of firms. We can define *employment* as a direct relational contract between an employer and a worker and *outsourcing* as an managed contract in which an intermediary firm employs workers and assigns them to different entrepreneurs. Under these definitions, the model can be used to predict the determinants and effects of professional service outsourcing.

Example (Global sourcing firms)**.** Consider a retailer (*buyer*) who needs a manufacturing service performed. It can fulfill the demand either by directly contracting with a manufacturer (*producer*) or contracting through a global sourcing firm (*manager*) such as Li & Fung. Such firms maintain a large number of relationships with upstream manufacturers and function as nexuses of relational contracts (Belavina and Girotra 2012). They also represent a substantial fraction of international trade.[13] We can define the service contract between the retailer and the manufacturer as a direct

[12]Professional service firms account for 12 percent of U.S. employment and are among the largest employers in the world (Berlingieri 2013). More broadly defined, services account for the vast majority of cross-establishment trade (Bostanci and Kambhampati 2022).

[13]Ahn, Khandelwal and Wei (2011) show that trade intermediaries account for around 20% of China's exports in 2005. Bernard, Grazzi and Tomasi (2015) document that more than one-quarter of Italian exporters are intermediaries, and they account for over 10% of exports.

contract and that between the retailer and the sourcing firm as a managed contract. Under these definitions, our model can be used to analyze the scope and effects of global sourcing firms.

3 Direct and Managed Relational Contracts

In this section we characterize optimal direct and managed relational contracts. We then analyze the buyer's choice between direct and managed contracts. This analysis explains why and when managed contracting may be preferred over direct contracting.

3.1 Direct Relational Contracts

To analyze direct relational contracts, we take the perspective of a single buyer i. Since producers are homogeneous, we omit the j subscript. We assume that the producer's pre-matching continuation value is exogenously given as $\overline{U} > 0$. We endogenize $\overline{U}$ in Section 4.

We say that strategies under a relational contract are *contract-specific* if they do not depend on the player's identity, calendar time, or any history outside the current contract. A relational contract is *stationary* if strategies are time-invariant functions of the buyer's demand realizations. A relational contract is offerer-optimal if it yields the highest possible surplus for the party offering the contract. We restrict our attention to offerer-optimal, contract-specific, stationary relational contracts in which producer's effort level is one if and only if $d_{it} = 1$.[14] These contracts must also satisfy the following two conditions. First, on the equilibrium path, parties within a match always choose to continue their relationship immediately after production. Second, off the equilibrium path, deviations are punished in the harshest possible way. These assumptions are standard in the literature (MacLeod and Malcomson, 1998; Baker, Gibbons and Murphy, 2002; Board and Meyer-ter Vehn, 2015).

[14]Even though there is limited liability in our model, as shown in Appendix Section B, focusing on stationary relational contracts is without loss since effort is perfectly observable and pairwise stability (or "bilateral efficiency" as in Board and Meyer-ter Vehn (2015)) is imposed on the equilibrium concept. A pairwise stable relational contract is a Pareto-optimal contract for parties in a match when they take their outside options as given. See Li (2022) for a more detailed discussion on how non-stationary relational contracts may be optimal in equilibrium when the pairwise stability restriction is relaxed.

Let $C_i^D = (w_{1i}, w_{0i}, \beta_i)$ denote a contract-specific, stationary relational contract offered by buyer i directly to a producer. In this contract, w_{1i} is the payment when $d_{it} = 1$, w_{0i} is the payment when $d_{it} = 0$, and $\beta_i \in [0, 1]$ is the probability that buyer i stays with the producer when the buyer's demand switches from one to zero. The time subscript is dropped since we focus on stationary contracts.

Under a direct contract, the buyer motivates the producer to exert effort using credible promises of future surplus from their contractual relationship. If the buyer deviates from the specified payment, the producer exerts no effort and separates from the buyer after production with probability one. If the producer deviates from the specified effort, the buyer separates from the producer after production with probability one. Since the buyer has fluctuating demand, the retention probability β_i determines the expected duration of the relationship and therefore affects the level of payments needed to incentive the producer.

If $\beta_i > 0$, the post-matching continuation payoffs for the producer when $d_{it} = 1$ and $d_{it} = 0$ are, respectively, given by

$$U_{1i} = w_{1i} - c + \delta\Big[(1 - \alpha_{1i})U_{1i} + \alpha_{1i}(\beta_i U_{0i} + (1 - \beta_i)\overline{U})\Big], \tag{1}$$

and

$$U_{0i} = w_{0i} + \delta\Big[\alpha_{0i}U_{1i} + (1 - \alpha_{0i})(\beta_i U_{0i} + (1 - \beta_i)\overline{U})\Big]. \tag{2}$$

The relevant incentive constraints for the producer are as follows:

$$U_{1i} \geq w_{1i} + \delta\overline{U}, \tag{P-IC-e}$$

$$U_{1i} \geq \overline{U}, \tag{P-IC1}$$

$$U_{0i} \geq \overline{U}. \tag{P-IC0}$$

Constraint (P-IC-e) requires the producer to choose effort over shirking when the service is needed. Constraints (P-IC1) and (P-IC0) require that the producer remain with the current buyer when the demand is 1 or 0, respectively. If $\beta_i = 0$, equation (2) and constraint (P-IC0) do not apply, as the buyer immediately separates from the producer when demand is zero.

For the buyer, the post-matching continuation payoffs when $d_{it} = 1$ and $d_{it} = 0$ are

$$\Pi_{1i} = y - w_{1i} + \delta\Big[(1 - \alpha_{1i})\Pi_{1i} + \alpha_{1i}(\beta_i \Pi_{0i} + (1 - \beta_i)\overline{\Pi}_{0i})\Big],$$

and

$$\Pi_{0i} = -w_{0i} + \delta\Big[\alpha_{0i}\Pi_{1i} + (1 - \alpha_{0i})(\beta_i \Pi_{0i} + (1 - \beta_i)\overline{\Pi}_{0i})\Big].$$

where $\overline{\Pi}_{1i}$ and $\overline{\Pi}_{0i}$ is the value of the buyer's pre-matching continuation values when $d_{it} = 1$ or $d_{it} = 0$, respectively. Since there is an excess of producers in the frictionless matching market, the buyer can always successfully find a match, so $\overline{\Pi}_{1i} = \Pi_{1i}$.

The relevant incentive constraint for the buyer are:

$$\Pi_{1i} \geq \delta(\alpha_{1i}\overline{\Pi}_{0i} + (1 - \alpha_{1i})\overline{\Pi}_{1i}), \tag{B-IC-w}$$

$$\Pi_{1i} \geq \overline{\Pi}_{1i}, \tag{B-IC1}$$

$$\Pi_{0i} \geq \overline{\Pi}_{0i}. \tag{B-IC0}$$

Constraint (B-IC-w) ensures that the buyer honors the payment to the producer. Constraints (B-IC1) and (B-IC0) reflect the buyer's desire to retain the producer when there is a demand or not, respectively. As before, if $\beta_i = 0$, the term Π_{0i} and constraint (B-IC0) do not apply, as the buyer would immediately separate from the producer.

The optimal direct contract is obtained by choosing w_{1i}, w_{0i}, and β_i to maximize Π_{1i}, subject to (B-IC-w), (B-IC1),(P-IC-e), (P-IC1), as well as (B-IC0) and (P-IC0) if $\beta_i > 0$.

Lemma 1. *Suppose an optimal direct relational contract exists. Under this contract, if*

$$\frac{(1 - \delta)\overline{U}}{c} > \frac{\alpha_{0i}}{1 - \alpha_{1i}}, \tag{3}$$

then the buyer pays

$$w_{DS}(\alpha_i) = \left(\frac{1}{\delta}\frac{1}{1 - \alpha_{1i}}\right)c + (1 - \delta)\overline{U}. \tag{4}$$

when demand is one and separates from the producer when demand becomes zero. Otherwise, the buyer pays

$$w_{DI}(\alpha_i) = \left(\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_{0i}}{(1-\alpha_{1i})+\frac{\delta}{1-\delta}\alpha_{0i}}\right)c + \left(\frac{1+\frac{\delta}{1-\delta}\alpha_{0i}}{(1-\alpha_{1i})+\frac{\delta}{1-\delta}\alpha_{0i}} - \delta\right)\overline{U} \tag{5}$$

when demand is one, remains matched with the idle producer when demands switches to zero, and pays zero until demand switches back to one.

Proof. All omitted proofs are in the appendix. □

Lemma 1 shows that the optimal direct contract features a buyer who either always separates from a producer or always retains him when their demand switches to zero. According to Equation (3), direct contracts with separation dominates direct contracts with idleness when (a) the producer's continuation value when unmatched $\overline{U}$ is high, (b) the buyer has a smaller α_{0i}, so a longer period of idleness is expected, and (c) the buyer has a larger α_{1i}, so a shorter period with service demand is expected.

The payment to producers in a direct contract, $w_D(\alpha_i) \equiv \min\{w_{DS}(\alpha_i), w_{DI}(\alpha_i)\}$, is increasing in α_{1i}. This is because when business needs are shorter-lived, the producer faces a higher chance of either separating or becoming idle, so the future surplus in the relationship is smaller. A higher incentive rent is therefore needed to incentivize producer effort.

3.2 Managed Relational Contracts

Under managed contracts, buyers delegate to managers the responsibility of motivating and monitoring producers. The manager fulfills the buyer's demand by entering relational contracts with a large number of producers and assigning producers to buyers according to demand realization.

We first analyze the contract that managers offer to producers. To meet buyer demand, we assume that each manager offers offerer-optimal, contract-specific, and stationary contracts to producers. As shown in Section 3.1, the terms in an optimal direct contract with producer hinge on the buyer's demand-switching probabilities. Unlike buyers, however, managers face constant demand for services and therefore have constant demand for effort from its matched producers. The reason is that each manager randomly matches with a continuum of buyers drawn from the same distribution, so by the law of large numbers, the managers face total demand from buyers that is constant over time. Anticipating this stable demand for services, the measure of producers that each manager contracts with is equal to the expected measure of demand realizations, and each matched producer is asked to exert effort in every period. Therefore, by the logic of Lemma 1, the compensating payment for the producer is $w_{1i} = w_M$ in every period, where

$$w_M = \frac{c}{\delta} + (1-\delta)\overline{U}. \tag{6}$$

We next consider how producers are assigned to buyers in each period under the managed contract. Note that buyers are indifferent between any assignment of producers where the assigned producer exerts effort, since producers are identical in our model. Producers are also indifferent between any assignment of buyers where the managers offer the same level of payments. Since managers require producers to exert effort and provide the same compensating payment in every period, buyers and producers have the same payoffs in any assignment where producers are matched with buyers with positive demand in every period. There are an infinite number of such assignments, and any such assignment is optimal.

We now characterize the contract that buyers offer to managers. Let $C_i^M = (p_{1i}, p_{0i}, \beta_i)$ denote a buyer-optimal, contract-specific, and stationary managed relational contract offered by a buyer to a manager. Here p_{1i} and p_{0i} are the service fees when the buyer needs and does not need the service, respectively. If the buyer deviates from the specified service fee, the manager does not assign a producer to the buyer and separates from the buyer. If instead the producer assigned by the manager deviates from the specified effort, the buyer separates from the manager after production. Since matching is not anonymous in the manager market, the buyer never chooses to match with the defaulted manager again in the future.[15]

[15]We assume that in the manager market, buyers randomly match with a manager who has not breached the relational contracts with any of them. Specifically, let $\mathcal{K}$ be the set of all managers and $\mathcal{K}_{i,t}$ be the subset who have interacted with buyer i and breached the relational contract with i before period t. Should a buyer become unmatched and match with a manager in the manager market in period t, she randomly matches with one of the managers from the set $\mathcal{K} \setminus \mathcal{K}_{i,t}$.

Under C_i^M, the post-matching continuation payoffs for the manager in a buyer-manager match, when the service is and is not needed, respectively, are

$$V_{1i} = p_{1i} - w_M + \delta\Big[(1-\alpha_{1i})V_{1i} + \alpha_{1i}(\beta_i V_{0i} + (1-\beta_i)\overline{V})\Big], \tag{7}$$

and

$$V_{0i} = p_{0i} + \delta\Big[\alpha_{0i}V_{1i} + (1-\alpha_{0i})(\beta_i V_{0i} + (1-\beta_i)\overline{V})\Big], \tag{8}$$

where $\overline{V}$ is a manager's continuation value after separating from a buyer. The relevant incentive constraints for the manager, similar with those for a producer, are

$$V_{1i} \geq p_{1i} + \delta\overline{V}, \tag{M-IC-w}$$

$$V_{1i} \geq \overline{V}, \tag{M-IC1}$$

$$V_{0i} \geq \overline{V}. \tag{M-IC0}$$

Note that $\overline{V} = 0$ on the equilibrium path. If a manager separates from a buyer, it cannot match with a new buyer, because all other potential buyers are matched with some manager and will not become unmatched on the equilibrium path.

For the buyer, the continuation payoffs and incentive constraints are the same as in the direct contract, except that the payments w_{0i} and w_{1i} to the producer are replaced with service fee payments p_{0i} and p_{1i} to the manager. For concision, we omit these conditions, which simply repeat (B-IC-w), (B-IC-1), and (B-IC0). The optimal managed contract maximizes Π_{1i} subject to these incentive compatibility constraints.

Lemma 2. *Suppose an optimal managed contract exists. Under this contract, the buyer always retains the manager, pays zero service fees to the manager when there is no demand, and when there is demand, she pays the manager a service fee equal to*

$$p(\alpha_i) = \left(\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_{0i}}{(1-\alpha_{1i})+\frac{\delta}{1-\delta}\alpha_{0i}}\right)w_M. \tag{9}$$

Lemma 2 shows that the cost of managed contract is a form of double marginalization. Note that $p(\alpha_i)$ can be rewritten as the product of $\lambda(\alpha_i) = \frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_{0i}}{(1-\alpha_{1i})+\frac{\delta}{1-\delta}\alpha_{0i}}$ and the payment to the producer w_M. Here $\lambda(\alpha_i)$ can be thought of as a manager markup. Furthermore, as shown in Equation (6), w_M is elevated above the cost of effort to the producer. In other words, both the manager and producer are paid rents so that both are incentivized to honor their contractual obligations.

The benefit of managed contract is that the payment to the producer is lower than the payment under direct contracts, since the manager can smooth demand across its buyers. To see this, note that $w_D(\alpha_i) \geq w_M$ for all α_i, where equality holds if and only if $\alpha_{1i} = 0$.

3.3 Optimal Contractual Choice

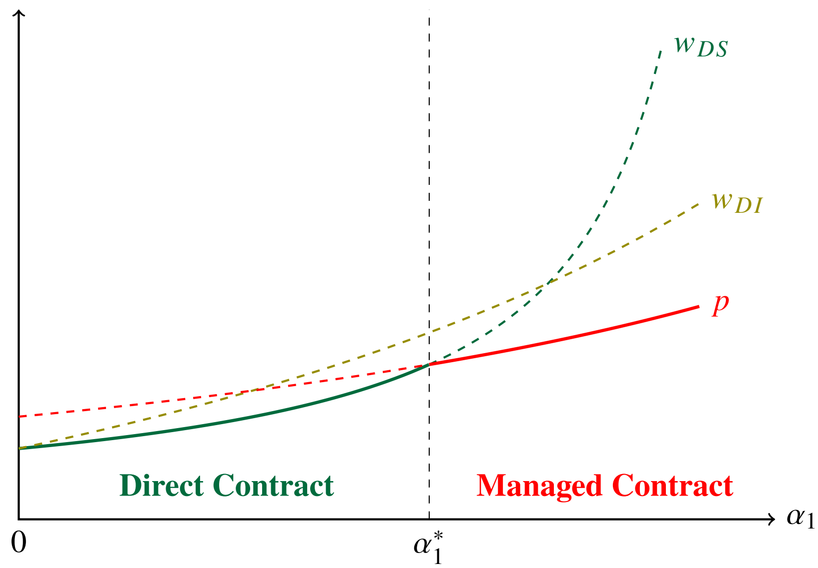
Having characterized direct and managed contracts, we now characterize when buyers choose managed contracts. We focus on buyers who have the same α_{0i}, which is the rate at which a buyer's demand changes from zero to one. We ask how the optimal contractual choice depends on α_{1i}, the rate at which the buyer's demand changes from one to zero, and $\overline{U}$, the producer's outside option, which in equilibrium reflects whether the producer market is tight and whether it is easy for producers to rematch.

To compare the two, it suffices to compare the payment $w_D(\alpha_i)$ under direct contracting and the service fee $p(\alpha_i)$ under managed contracting. This is because the buyer pays nothing when their demand is zero and can always match with a producer when her demand is one.

Figure 3 provides a graphical illustration of this comparison. For a buyer with stable demand, i.e., for whom $\alpha_{1i} = 0$, managed contracting is strictly more expensive than direct contracting because of double marginalization. To obtain high-quality services, the buyer needs to pay additional rent to the manager. However, there is no benefit to managed contracts, since demand is stable, so the producer is paid the same incentive rent under direct contracting.

The advantage of managed contracting over direct contracting becomes larger when business needs are short-term. As α_{1i} becomes larger, producers must be paid elevated payments in order for them to exert effort. Since managers can reassign producers across buyers based on

Figure 3: Service cost under direct and managed contracts



Note: α_1 is the probability that the buyer's demand switches from 1 to 0.

demands so producers neither separate nor become idle, the cost of incentivizing producers is lowered. To see this mathematically, note that $w_{DS}(\alpha_i)$ approaches infinity as α_{1i} approaches one. Furthermore, $p(\alpha_i)$ increases in α_{1i} less steeply than $w_{DI}(\alpha_i)$ if c is relatively small and $\overline{U}$ is relatively large. Therefore, when rematching is easy, the manager operates a more cost-efficient internal producer market by having long-standing relational contracts on both sides of the market.

Figure 4 graphically shows the optimal contracts as a function of α_{1i} and $\overline{U}$. In this figure, provided $\overline{U} > \overline{U}^*$, then there exists a cutoff value such that managed contracts dominates if and only if α_{1i} is sufficiently large. If $\overline{U} < \overline{U}^*$, then the producer's pre-matching continuation value is low, so it is optimal to retain them and keep them idle until demand returns.

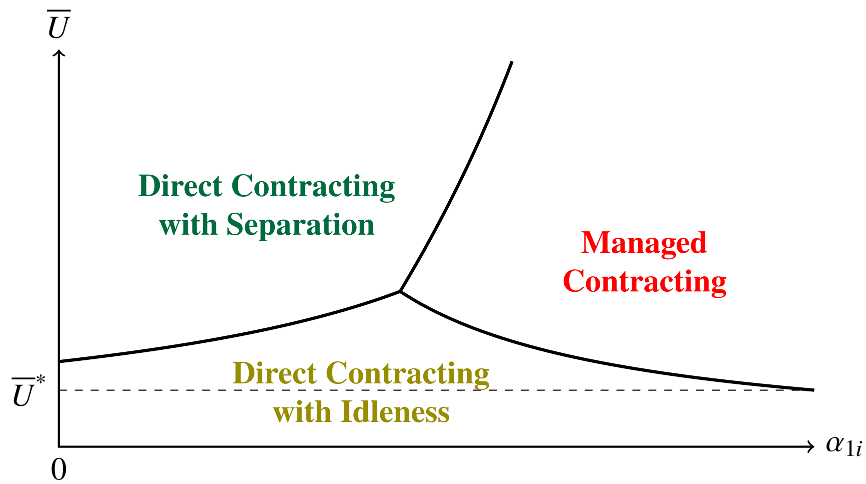
The following proposition formalizes this result.

Proposition 1. *Take any set of buyers i with the same α_{0i} for whom an optimal contract exists. There exists $\overline{U}^* > 0$ such that:*

1. *If $\overline{U} < \overline{U}^*$, a direct contract is optimal for all i in this set;*

2. *If $\overline{U} > \overline{U}^*$, there exists $\alpha_1^*(\alpha_{0i}, \overline{U}) \in (0, 1)$ such that a managed contract is optimal if and only if $\alpha_{1i} \geq \alpha_1^*(\alpha_{0i}, \overline{U})$.*

Figure 4: Optimal contractual choice given model parameters



Proposition 1 can be viewed the answer to a variant of the question first posed by Coase (1937): Why and under what conditions should we expect centralized allocators to emerge in decentralized markets? The above result shows that managerial coordination dominates direct pairings when business needs are short-term and the continuation value of an unmatched producer is high. Moreover, managerial coordination can dominate bilateral pairings even though buyers and producers in the model can frictionlessly meet, possess no hidden information about capabilities, and can freely enter contracts. Our result therefore implies that centralized allocators may emerge in the absence of many types of transaction costs, including search, bargaining, and contract-writing costs that are previously emphasized in the theory of the firm.

The model's prediction that buyers with frequent demand are more likely to disintermediate is consistent with findings in related literature. Williamson (1985) attributes this tendency to administrative and bargaining costs in repeated transactions. Wernerfelt (2015, 2016) attributes it to switching costs. In our model, the cost of managed contracts is instead a price premium charged by the manager to ensure its performance as an aggregator, monitor, and contract enforcer.

4 Steady-State Equilibrium

In this section, we show that there exists a unique steady-state equilibrium. We then explore the model's empirical implications. First, we compare the outcomes of managed producers and directly contracted producers in equilibrium. Second, we compare the outcomes and welfare of buyers and producers in economies with and without managers. Third, we study how contractual choices depend on market size. We then discuss market-level spillover effects from managerial coordination.

4.1 Deriving the Equilibrium

We say that the economy is in a steady state when (1) the number of producers in the economy and the distribution of contracts in the matching market are unchanging across periods and (2) each buyer's demand evolves according to its stationary distribution. By the properties of Markov processes, the steady-state probability that a buyer i has positive demand in any period is equal to $\pi_i = \alpha_{0i}/(\alpha_{0i} + \alpha_{1i})$.

At the start of each period, some buyers and managers are unmatched and directly offer direct contracts to unmatched producers. Let $\mathcal{I}_{DS}$ denote the set of buyers who enter direct contracts that end when their demand switches to zero. For a buyer $i \in \mathcal{I}_{DS}$, the steady-state probability that they offer new contracts is

$$v_i = (1 - \pi_i)\alpha_{0i}. \tag{10}$$

Let $\mathcal{I}_{DI}$ be the set of buyers who directly contract with producers in contracts that continue when demand switches. Let $\mathcal{I}_M$ be the set of buyers who contract with managers. For both types of contracts, producers never separate. Therefore, for any buyer $i \in \mathcal{I}_{DI} \cup \mathcal{I}_M$, the steady-state probability that they offer new contracts is $v_i = 0$. The total measure of new contracts offered by buyers and managers in the producer market is given by

$$v = \int_{\mathcal{I}_{DS}} v_i dF. \tag{11}$$

The steady-state measure of producers in direct contracts, n_B, is given by

$$n_B = \int_{\mathcal{I}_{DS}} \pi_i dF + \int_{\mathcal{I}_{DI}} dF. \tag{12}$$

The steady-state measure of producers in managed contracts, n_M, is given by

$$n_M = \int_{\mathcal{I}_M} \pi_i dF. \tag{13}$$

The measure of unmatched producers after matching is

$$n_N = n - n_B - n_M. \tag{14}$$

If y and C are sufficiently large and the buyer type distribution F has full support on $[0,1] \times [0,1]$, the value of entering either direct or managed contracts is higher than the value of being unmatched. This implies that there is an excess of producers who enter the producer market, so $n_N > 0$.

Since matching is frictionless, all contracts offered in the producer market are immediately filled. The total measure of unmatched producers before matching, u, is given by

$$u = v + n_N. \tag{15}$$

Matching is random and contract offers are never made to matched producers, so the Bellman equation for an unmatched producer is given by

$$\overline{U} = \int_{\mathcal{I}_{DS}} \frac{v_i}{u} U_{DS}(\alpha_i) dF + \left(1 - \frac{v}{u}\right) \overline{U}, \tag{16}$$

where $U_{DS}(\alpha_i) = \frac{1}{1-\delta(1-\alpha_{1i})}(w_{DS}(\alpha_i) - c + \delta\alpha_{1i}\overline{U})$. Note that the continuation value of being unmatched ($\overline{U}$) consists of two components. The first term reflects the rent from matching with an unmatched buyer. The second term reflects the value of remaining unmatched.

To close the model, we solve for the steady-state number of producers that enter the economy.

By assumption, producers enter the economy at the beginning of each period by paying an entry cost C. Entry drives down the likelihood that producers are matched, so they enter only until the continuation value of being unmatched in the labor market equals their entry cost. This yields the following condition:

$$\overline{U} = C. \tag{17}$$

We can now define an equilibrium in our economy.

Definition 1. *A steady-state equilibrium is a distribution of relational contracts offered by each buyer and manager such that:*

1. *All relational contracts are offerer-optimal, contract-specific, and stationary;*
2. *Each player's pre-matching continuation value is determined by steady-state transition probabilities and frictionless and random matching via Equation (16);*
3. *The measure of producers in the economy is derived from the producer entry condition, given by Equation (17).*

Proposition 2. *There exists a unique steady-state equilibrium. If y and C are sufficiently high and F has full support on $[0,1] \times [0,1]$, then a non-zero measure of buyers enter direct contracts, while another non-zero measure of buyers enter managed contracts.*

Proof. By Equation (17), $\overline{U}$ equals the entry cost C. Given $\overline{U}$, we can compare the values of $w_{DS}(\alpha_i)$, $w_{DI}(\alpha_i)$, $p(\alpha_i)$, and y for each i using Equations (4), (5), and (9) to determine $\mathcal{I}_{DS}$, $\mathcal{I}_{DI}$, and $\mathcal{I}_M$. Having derived these, we can obtain unique values for v_i, v, n_B, and n_M from Equations (10), (11), (12), and (13). We plug these into Equation (16) to solve for a unique value for u. Plugging u into Equations (14) and (15) then yields a unique value for n. The desired statement then follows from Proposition 1. □

4.2 Empirical Implications

Having shown that there exists a unique steady-state equilibrium in this exchange economy, we can now explore the model's predictions regarding equilibrium behavior.

Effects of Managers on Producers

How does managerial coordination affect the pay, separation rates, and idleness of producers? Empirical studies have found that workers managed by professional service firms earn lower and more compressed wages (Dube and Kaplan 2010; Goldschmidt and Schmieder 2017; Drenik et al. 2020). The employees of professional service firms are also documented to have lower hazard into unemployment than comparable direct employees (Guo, Li and Wong 2024). Similar patterns are found in our model. We focus on the interesting case where all contract types coexist and derive four findings.[16]

First, directly contracted producers in our model have *higher average pay* per unit effort than managed producers in the steady-state equilibrium. This follows from the fact that $w_D(\alpha_i) > w_M$ for all buyers i with fluctuating demand.

Second, directly contracted producers have *more dispersed pay* per unit effort than those of managed producers. This is because $w_D(\alpha_i)$ takes on different values depending on α_i, which is heterogeneous across buyers, while w_M is constant.

Third, directly contracted producers have *higher separation rates*. We define a producer's separation rate as the probability that a matched producer becomes unmatched at the start of the next period. This probability is higher for directly contracted producers, since they may separate from buyers when their demand changes, while managed producers are always reallocated among the manager's clients.

Fourth, directly contracted producers are *more likely to be idle* during their employment spells. We define idleness as the steady-state probability that a producer is matched but does not exert effort. In our model, managed producers are never idle, while directly contracted producers may have positive idleness.

Corollary 1. *Compared to managed producers, directly contracted producers have higher average pay, more dispersed pay, higher separation rates, and higher idleness in equilibrium.*

[16] Specifically, we assume that y, C and F are such that $|\mathcal{I}_M|, |\mathcal{I}_{DI}|, |\mathcal{I}_{DS}| > 0$. If F has full support on $[0, 1] \times [0, 1]$, then by Proposition 1 there exists y and C such that this holds.

Managerial Coordination is Limited by the Extent of the Market

How does market size affect the extent of managerial coordination? Felix and Wong (2024) document that the share of security guards managed by security firms is higher in larger Brazilian microregions. Similarly, in our model, managerial coordination is efficiency-enhancing only if there are multiple sellers and buyers. If there is only one buyer and one producer in the economy, it is never profitable for a buyer to contract with an manager, since the manager will require a markup and there are no benefits from demand aggregation. Therefore, managerial coordination is more likely when *the extent of the market*—i.e. the available number of buyers and sellers—is large. This result is closely related to Stigler (1951), who conjectures that firms spin-off production stages because of increasing economies of scale as the market grows.

Remark 1. *In an economy with a single buyer and a single producer, managers are always matched with zero buyers and zero producers.*

Trade and Welfare with and without Managers

How does the presence of managers alters market equilibrium and the welfare of buyers and producers? Felix and Wong (2024) show that Brazil's 1993 legalization of non-core activity outsourcing sharply increased the use of security firms, persistently increased security guard employment, and displaced incumbent security guards from high-wage direct employers.

Similar patterns can arise in our model. To show this, we consider two economies: one with managers and one without. We assume that producers freely enter at some exogenous cost C in both economies. Therefore, the producer's continuation values when unmatched, and hence the contractual terms offered to them by buyers under direct contracting, are the same in the two economies. The only difference is the added possibility of managed contracts.

The introduction of managers can cause three types of buyers to switch to managerial coordination: buyers who do not initially consume services, buyers who initially choose direct contracts with idleness, and buyers who initially choose direct contracts with separation. We denote the three subsets of switching buyers as $\mathcal{S}_0$, $\mathcal{S}_I$, and $\mathcal{S}_S$.[17] The contracting choices of the

[17]Assuming that indifferent buyers do not switch, we have that $\mathcal{S}_0 = \{i \mid p(\alpha_i) < y < w_D(\alpha_i)\}$, $\mathcal{S}_I = \{i \mid$

remaining buyers are unchanged. We focus on the case where the measure of buyers switch from no consumption to managerial coordination and the measure of buyers switch from direct contracts to managerial coordination are both nonzero.[18]

We find that the introduction of managers benefits buyers with fluctuating demand, but it hurts incumbent producers who were initially earning pay premiums. By Corollary 1, the average producer pay per unit effort falls. Producer separation rates and idleness also fall. At the same time, the per-period payoff of buyers who switch to managerial coordination increases, while the per-period payoff of the remaining buyers are unchanged.

The introduction of managers also unambiguously increases the measure of buyers who receive services. However, it is ambiguous whether the measure of matched producers ($n_M + n_B$) increases or falls. On one hand, managers enable more buyers to afford services, which increases demand for producers. On the other, managers reduce idleness, so fewer producers are needed to fulfill the same level of demand. The relative magnitude of these two effects depends on the distribution of buyer types.

Corollary 2. *When managers are introduced into an economy, the payoffs of some buyers increase, while the payoffs of the remaining buyers remain unchanged. The measure of buyers who receive services increases. The mean pay per unit effort received by producers falls. The total measure of matched producers increases if and only if* $\int_{\mathcal{S}_0} \pi_i dF > \int_{\mathcal{S}_I} (1 - \pi_i) dF$.

4.3 Exogenous Producer Population

So far we have assumed that the number of producers is determined via endogenous entry. This assumption rules out indirect market-level spillover effects. In this section, we analyze an economy where the number of producers is fixed exogenously at $n > 1$.

We first show that if n is sufficiently large, then the producer market becomes very slack, buyers prefer to enter contracts with idleness, and there is no managerial coordination.

$p(\alpha_i) < w_{DI}(\alpha_i) < \min\{y, w_{DS}(\alpha_i)\}\}$, and $\mathcal{S}_S = \{i \mid p(\alpha_i) < w_{DS}(\alpha_i) < \min\{y, w_{DI}(\alpha_i)\}\}$.

[18]Formally, we assume that y, C and F are such that $|\mathcal{S}_0| > 0$ and $|\mathcal{S}_I \cup \mathcal{S}_S| > 0$. This condition requires that there exists $\alpha_i, \alpha_i' \in \operatorname{supp} F$ such that $y > w_D(\alpha_i) > p(\alpha_i)$ and $w_D(\alpha_i') > y > p(\alpha_i')$. If F has full support on $[0, 1] \times [0, 1]$, then by Proposition 1 there exist y and C such that this holds.

Proposition 3. *Assume the number of producers n is exogenously fixed. There exists a unique steady-state equilibrium. If n is sufficiently large, then there are no managed contracts.*

Proof. Given $\overline{U}$, we can compare the values of $w_{DS}(\alpha_i)$, $w_{DI}(\alpha_i)$, $p(\alpha_i)$, and y for each i using (4), (5), and (9) to determine $\mathcal{I}_{DS}$, $\mathcal{I}_{DI}$, and $\mathcal{I}_M$. Having derived these, we can obtain unique values for v_i, v, n_B, and n_M from (10), (11), (12), and (13). Plugging $w_{DS}(\alpha_i)$, $w_{DI}(\alpha_i)$, and $w_M(\alpha_i)$ into (16) yields a contraction mapping. Therefore a unique fixed point for $\overline{U}$ exists. By (14), (15), and (16), $\overline{U}$ is decreasing in n, and $\overline{U} \to 0$ as $n \to \infty$. The desired result follows by Proposition 1. □

We next compare economies with and without managers assuming that n is fixed and that some nonzero measure of buyers choose managerial coordination when possible. In this economy, introducing managers has both direct and indirect effects. The direct effect holds $\overline{U}$ fixed and was characterized in the previous subsection. Due to this effect, various types of buyers switch to managerial coordination, producer rents fall, and the number of matched producers may or may not increase. Without free entry, however, $\overline{U}$ also adjusts. A change in $\overline{U}$ affects all of the endogenous variables in the model, so managerial coordination has indirect spillover effects onto all producers.

The sign of the resulting change in $\overline{U}$ depends on the distribution of buyer types α_i. If a large set of buyers switch from being unmatched to matched, then $\overline{U}$ would increase due to reduced wait time in a tightened producer market. This indirect effect raises the pay of all producers. On the other hand, if no buyers switch from being unmatched to being matched, then $\overline{U}$ would unambiguously fall due to producer rent reductions associated with initially matched buyers switching to managerial coordination. This indirect effect instead reduces the pay of all producers.

5 Extensions

This section extends the model to study the relationship between managerial coordination, producer specialization, and manager reputation concerns. We first incorporate an endogenous choice by producers to specialize in different capabilities upon entry into the economy. Then, we add the possibility that poor performance by an manager is made known to a broader set of buyers.

5.1 Specialization

We consider a multi-task economy with a measure of buyers who have unit demand in every period, for either one of two tasks $d_{it} \in \{A, B\}$. Each buyer's demand switches from one task to another task with some symmetric probability α_i at the start of each period.[19] There is also an excess measure of producers who choose whether to become either specialists in one of two activities needed by buyers or a generalist with middling skill in both services. Let $\phi_j \in \{A, B, G\}$ denote the chosen type of the producer, where A and B refer to specialists and G refers to the generalist. The output depends on the buyer's demand d_{it}, the producer's type ϕ_j, and the producer's chosen effort e_t, and is given by

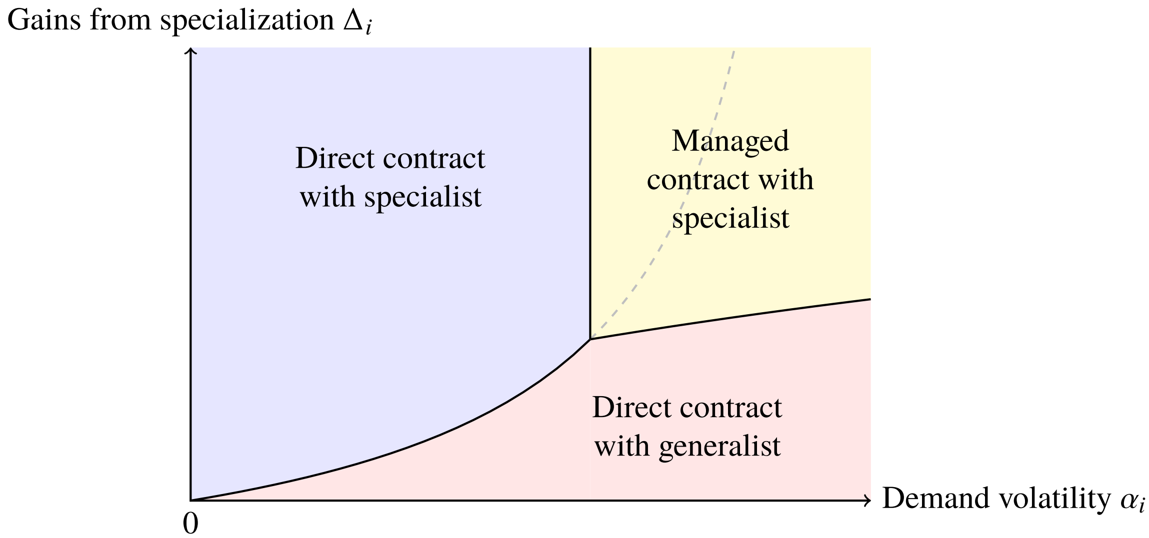
$$y_{it} = \left[y \cdot \mathbf{1}\{\phi_j = G\} + (y + \Delta_i) \cdot \mathbf{1}\{\phi_j = d_{it}\}\right] e_t,$$

where $\Delta_i > 0$ denotes the buyer-specific *gains from specialization*. If the specialist exerts effort, output is high when demand and the producer's type are well-matched, but low when they are not. Output is always middling for generalists who exert effort.

As before, neither pay w_t nor effort e_t are contractible and must be incentivized through relational contracts. We assume that output y is sufficiently large so that buyers are always able to receive positive profit by directly contracting with a generalist, so there are never buyers who do not receive services. We also assume that each producer's entry cost C is sufficiently large so that specialist producers never remain in a contract but become idle when the demand of their buyer changes. These assumptions allow us to focus on each buyer's choice between direct contracting with a generalist, direct contracting with a specialist who is never idle, and managed contracting. There are K managers for each task. Each manager can only monitor producers

[19] This is essentially simplifying the model in Section 2 by assuming $\alpha_{Ai} = \alpha_{Bi} = \alpha_i$.

Figure 5: Optimal contractual choice in the presence of gains from specialization



specializing in that task and are randomly matched with buyers who offer managed contracts.[20]

Drivers of Managerial Coordination in a Multi-task Economy

Which buyers choose managed contracts in a multi-task economy? Abraham and Taylor (1996) find that establishments with cyclical demand and specialized needs are more likely to outsource accounting services to external firms. Similar patterns are found in our model.

In the model, managed contracts dominates direct contracts for buyers with high demand volatility and large gains from specialization, as shown in Figure 5. Direct contracts are optimal otherwise, due to double marginalization. It is never optimal for a buyer to enter managed contracts with generalist producers. In equilibrium, some buyers choose managed contracts with specialists, while others directly contract with either specialists or generalists.

Proposition 4. *A unique steady-state equilibrium exists in a multi-task economy. Buyer i chooses managed contracts if and only if their demand volatility α_i and gains from specialization Δ_i are both sufficiently large.*

[20] This setup implicitly assumes that each manager specializes in monitoring one type of task.

Division of Labor with or without Managers

How does the presence of managers affect the division of labor? Garicano and Hubbard (2009) find that the share of lawyers that specialize and the share of lawyers working in specialized firms both increase with market size.[21]

Similarly, in our model there are no managers when the market is too small (Remark 1). Without managers, buyers with more volatile demand contract with generalists, while those with less volatile demand contract with specialists. The dashed curve in Figure 5 shows the boundary between direct contracting with a specialist and with a generalist.

When managerial coordination becomes possible, part of the demand for directly contracted generalists is replaced with demand for managed specialists. This causes the overall demand for specialists to rise. In response, more producers choose to become specialists. Correspondingly, there are fewer direct contracts with elevated pay, so fewer producers enter, and the measure of unmatched producers falls.

To show this formally, we assume that a non-zero measure of buyers choose managed contracts and the conditional distribution of α_i given Δ_i has positive support on $[0, 1]$.

Corollary 3. *When managers are introduced into a multi-task economy, the measure of specialist producers increase and the measure of unmatched producers falls.*

5.2 Manager Reputational Concerns

Thus far, our model follows Shapiro and Stiglitz (1984) in assuming that producers are motivated to perform through the threat of contract termination, which would require them to wait to rematch in an anonymous matching market. Another source of motivation, as modeled by Klein and Leffler (1981), is that producers may lose reputational capital when they renege on promises to deliver high-quality services, causing a broader set of buyers to withhold future business.

In this subsection, we argue that reputation concerns can play an important role in shaping the choice between managed and direct contracting. Specifically, managerial coordination becomes cheaper if the performance of the manager is partially observable to outside parties.

[21] For related evidence, see also Baumgardner (1988*a*,*b*) and Duranton and Jayet (2011).

To see this formally, we consider a multi-task economy with reputable managers, where low effort by a producer is communicated with some probability to other buyers who can withhold future business from the producer's matched manager. Specifically, we assume that with probability $\gamma \in [0, 1]$, the effort choice by a producer contracted by the manager is observed by another buyer, who is drawn among all buyers with uniform probability.[22] The parameter γ measures the *ease of word-of-mouth communication* about manager performance enabled by communication technologies such as the Internet and social media platforms.

Communication Technology, Manager Coordination, and the Division of Labor

How does the arrival of communication technologies like the Internet and social media affect the extent of managerial coordination and the division of labor? Bergeaud et al. (2021) provide causal evidence that the rise of broadband internet increased the employment share of professional service firm and the average occupational concentration of firms. Macroeconomic studies provide correlational evidence that as the employment share of professional service firms has increased, workers and firms have become increasingly specialized (Song et al. 2019; Handwerker 2023), and unemployment has fallen (Katz and Krueger 1999).

These facts can be interpreted in our model as follows. As the ease of communication increases, the manager faces a harsher punishment if it reneges on its contracts. To see this, note that an manager's mean continuation value from being matched with an buyer is $\tilde{V} = \frac{1}{|\mathcal{I}|} \int_{\mathcal{I}_O} \pi_i V_{1i} + (1 - \pi_i) V_{0i} dF > 0$. When $\gamma > 0$, the manager's binding IC constraint is now given by:

$$V_1 \geq p_1 + \delta(-\gamma \tilde{V}), \tag{M-IC-w'}$$

In other words, the threat of multilateral punishment means that a reduced mark-up is needed to incentivize the manager to perform. The unit cost of managed service therefore falls as γ increases.

As the cost of managerial coordination decreases, a larger measure of buyers choose managerial

[22]Here the manager will not wish to renege on more than one of its clients if she does not wish to do so for a single client, since an buyer may learn of bad service provided to multiple clients from word-of-mouth communication but can only punish maximally once.

coordination. Correspondingly, the measure of buyers who directly contract with specialists and generalists both fall. This enables greater producer specialization, since total demand for specialized producers increases. At the same time, there are fewer direct contracts with elevated pay, so the measure of unmatched producers falls.

Proposition 5. *A unique steady-state equilibrium exists in a multi-task economy with reputable managers. As the ease of word-of-mouth communication increases, the measure of managed producers increases, the measure of specialist producers increases, and the measure of unmatched producers falls.*

Remark. By construction, our model disallows two possibilities that may in reality affect the choice between managerial coordination and decentralized exchange. First, producers themselves may build and maintain reputational capital. If so, they would not rematch on an *anonymous* market, as modeled above. Instead, if a producer shirks their contractual responsibilities, buyers can cause producers to face difficulty rematching thereafter. In this case, the producer faces a stronger incentive to perform and does so even if the particular buyer they currently provide services for is expected to have no demand next period. Therefore, managerial coordination becomes less attractive if producers maintain reputational capital.

Second, producers matched with the manager may be able to communicate with one another about the manager's actions, such as through regular business conferences. This opens up the possibility that producers collectively punish managers in response to contract infringement (a la Levin 2002). In our model, the threat of losing multiple producer relationships has no bite because managers can immediately rematch with new producers. However, if rematching with producers is assumed to be difficult for managers, then multilateral relational contracts between an manager and its producers will increase the manager's incentive to perform, thereby lowering its service fee. Managerial coordination therefore becomes more attractive if producers can collectively punish managers.

6 Conclusion

This paper develops a general equilibrium model of an exchange economy in which managers endogenously emerge as centralized monitor-coordinators. In the model, managers function as nexuses of relational contracts and ease a fundamental tension between incentive provision and flexible allocations. The model highlights that the use of managers is shaped by a trade-off between demand aggregation and double marginalization. Because of this trade-off, managers only partially replace anonymous matching markets in determining assignments between buyers and producers. We solve for a unique steady-state equilibrium in economies with either free entry of producers or an exogenous producer population. We show how buyers sort into managerial coordination and decentralized exchange depending on their demand volatility, gains from specialization, market tightness, market size, and communication frictions. We characterize the effects of managers on output, the patterns of specialization, and the distribution of economic rents. We also discuss connections to the empirical literature on professional service firms.

A strength of our model is its parsimony. We show that centralized monitor-coordinators can emerge in exchange economies even in the absence of search frictions, adverse selection, production complementarities, market-level reputation effects, as well as reduced-form contract-writing and bargaining costs. However, our theory of managerial coordination ignores many first-order issues in the study of economic organizations, such as the scope of activities monitored by managers, the monitoring span of managers, multi-layered hierarchical structures, promotional ladders, the allocation of ownership rights, and the flow of information. As such, our model is directly applicable only to rudimentary organizations, such as professional service firms with identical workers. It is not hard, though, to imagine that general equilibrium models of relational contracting networks can be extended to shed light on many of the above issues. Further development of such models may therefore be a promising avenue for advancing the study of firm-like organizations.

References

Abraham, Katharine G. and Susan K. Taylor. 1996. "Firms' use of outside contractors: Theory and evidence." *Journal of Labor Economics* 14(3):394–424.

Ahn, JaeBin, Amit K Khandelwal and Shang-Jin Wei. 2011. "The role of intermediaries in facilitating trade." *Journal of International Economics* 84(1):73–85.

Alchian, Armen A. and Harold Demsetz. 1972. "Production, information costs, and economic organization." *The American Economic Review* 62(5):777–795.

Andrews, Isaiah and Daniel Barron. 2016. "The allocation of future business: Dynamic relational contracts with multiple agents." *American Economic Review* 106(9):2742–59.

Baker, George, Robert Gibbons and Kevin J Murphy. 2002. "Relational contracts and the theory of the firm." *Quarterly Journal of Economics* 117(1):39–84.

Bardhan, Pranab, Dilip Mookherjee and Masatoshi Tsumagari. 2013. "Middlemen margins and globalization." *American Economic Journal: Microeconomics* 5(4):81–119.

Baumgardner, James R. 1988*a*. "The division of labor, local markets, and worker organization." *Journal of Political Economy* 96(3):509–527.

Baumgardner, James R. 1988*b*. "Physicians' services and the division of labor across local markets." *Journal of Political Economy* 96(5):948–982.

Belavina, Elena and Karan Girotra. 2012. "The relational advantages of intermediation." *Management Science* 58(9):1614–1631.

Bergeaud, Antonin, Clement Malgouyres, Clement Mazet-Sonilhac and Sara Signorelli. 2021. Technological change and domestic outsourcing. CEP Discussion Papers dp1784 Centre for Economic Performance, LSE.

Berlingieri, Giuseppe. 2013. "Outsourcing and the rise in services." *Working Paper* .

Bernard, Andrew, Marco Grazzi and Chiara Tomasi. 2015. "Intermediaries in International Trade: Products and Destinations." *The Review of Economics and Statistics* 97(4):916–920.

Biglaiser, Gary and James W. Friedman. 1994. "Middlemen as guarantors of quality." *International Journal of Industrial Organization* 12(4):509–531.

Board, Simon. 2011. "Relational Contracts and the Value of Loyalty." *American Economic Review* 101(7):3349–67.

Board, Simon and Moritz Meyer-ter Vehn. 2015. "Relational contracts in competitive labour markets." *Review of Economic Studies* 82(2):490–534.

Bolton, Patrick and Ashvin Rajan. 2003. "The Employment Relation and the Theory of the Firm: Arm's Length Contracting vs Authority." *Working Paper* .

Bostanci, Gorkem and Ashwin Kambhampati. 2022. "A theory of firm boundaries with long-run incentives and at-will employment." *Working Paper* .

Budish, Eric. 2023. Trust at scale: The economic limits of cryptocurrencies and blockchains. Technical report Working Paper, U. of Chicago.

Calvert, Randall. 1995. Rational actors, equilibrium, and social institutions. In *Explaining Social Institutions*, ed. Jack Knight and Itai Sened. University of Michigan Press pp. 57–93.

Coase, Ronald H. 1937. "The nature of the firm." *Economica* 4(16):386–405.

Demsetz, Harold. 1988. "The theory of the firm revisited." *Journal of Law, Economics, & Organization* 4(1):141–161.

Drenik, Andres, Simon Jäger, Miguel Pascuel Plotkin and Benjamin Schoefer. 2020. "Paying outsourced labor: Direct evidence from linked temp agency-worker-client data." *Working Paper* .

Dube, Arindrajit and Ethan Kaplan. 2010. "Does outsourcing reduce wages in the low-wage service occupations? Evidence from janitors and guards." *ILR Review* 63(2):287–306.

Duranton, Gilles and Hubert Jayet. 2011. "Is the division of labour limited by the extent of the market? Evidence from French cities." *Journal of Urban Economics* 69(1):56–71.

Dyer, Jeffrey H. and Harbir Singh. 1998. "The relational view: Cooperative strategy and sources of interorganizational competitive advantage." *The Academy of Management Review* 23(4):660–679.

Fafchamps, Marcel. 2004. *Market institutions in sub-Saharan Africa*. Cambridge, MA: MIT Press.

Fahn, Matthias. 2017. "Minimum wages and relational contracts." *The Journal of Law, Economics, and Organization* 33(2):301–331.

Fahn, Matthias and Takeshi Murooka. 2022. Informal Incentives and Labor Markets. CESifo Working Paper Series 9740 CESifo.

Felix, Mayara and Michael B. Wong. 2024. "The Reallocation Effects of Domestic Outsourcing." *Working Paper* .

Fong, Yuk-fai and Jin Li. 2017. "Relational contracts, limited liability, and employment dynamic." *Journal of Economic Theory* 169:270–293.

Garicano, Luis and Thomas N Hubbard. 2009. "Specialization, firms, and markets: The division of labor within and between law firms." *The Journal of Law, Economics, & Organization* 25(2):339–371.

Geertz, Clifford. 1962. "The rotating credit association: A "middle rung" in development." *Economic Development and Cultural Change* 10(3):241–263.

Geertz, Clifford. 1978. "The bazaar economy: Information and search in peasant marketing." *American Economic Review* 68(2):28–32.

Gibbons, Robert. 2005. "Four formal (izable) theories of the firm?" *Journal of Economic Behavior & Organization* 58(2):200–245.

Gibbons, Robert and Rebecca Henderson. 2012. "Relational contracts and organizational capabilities." *Organization Science* 23(5):1350–1364.

Gil, Ricard and Giorgio Zanarone. 2017. "Formal and Informal Contracting: Theory and Evidence." *Annual Review of Law and Social Science* 13(1):141–159.

Goldschmidt, Deborah and Johannes F. Schmieder. 2017. "The rise of domestic outsourcing and the evolution of the German wage structure." *The Quarterly Journal of Economics* 132(3):1165–1217.

Granovetter, Mark. 1985. "Economic action and social structure: The problem of embeddedness." *American Journal of Sociology* 91(3):481–510.

Greif, Avner. 1993. "Contract enforceability and economic institutions in early trade: The Maghribi traders' coalition." *The American economic review* pp. 525–548.

Greif, Avner. 2006. *Institutions and the Path to the Modern Economy: Lessons from Medieval Trade*. Political Economy of Institutions and Decisions Cambridge University Press.

Greif, Avner, Paul Milgrom and Barry R Weingast. 1994. "Coordination, commitment, and enforcement: The case of the merchant guild." *Journal of political economy* 102(4):745–776.

Grossman, Sanford J. and Oliver D. Hart. 1986. "The costs and benefits of ownership: A theory of vertical and lateral integration." *Journal of Political Economy* 94(4):691–719.

Guo, Naijia, Duoxi Li and Michael B. Wong. 2024. "Domestic Outsourcing and Employment Security." *Working Paper* .

Handwerker, Elizabeth Weber. 2023. "Outsourcing, Occupationally Homogeneous Employers, and Wage Inequality in the United States." *Journal of Labor Economics* .

Hart, Oliver and John Moore. 2008. "Contracts as reference points." *The Quarterly Journal of Economics* 123(1):1–48.

Henderson, M.T. and S. Churi. 2019. *The Trust Revolution*. Cambridge University Press.

Jensen, Michael C. and William H. Meckling. 1976. "Theory of the firm: Managerial behavior, agency costs and ownership structure." *Journal of Financial Economics* 3(4):305–360.

Katz, Lawrence F. and Alan B. Krueger. 1999. "The High-Pressure U.S. Labor Market of the 1990s." *Brookings Papers on Economic Activity* 30(1):1–88.

Klein, Benjamin and Keith B. Leffler. 1981. "The role of market forces in assuring contractual performance." *Journal of Political Economy* 89(4):615–641.

Levin, Jonathan. 2002. "Multilateral contracting and the employment relationship." *The Quarterly Journal of Economics* 117(3):1075–1103.

Levin, Jonathan. 2003. "Relational incentive contracts." *American Economic Review* 93(3):835–857.

Levin, Jonathan and Steven Tadelis. 2010. "Contracting for government services: Theory and evidence from US cities." *The journal of industrial economics* 58(3):507–541.

Li, Duoxi. 2022. "Relationship Building under Market Pressure." *Available at SSRN 4380443* .

Li, Jin and Michael Powell. 2020. "Multilateral interactions improve cooperation under random fluctuations." *Games and Economic Behavior* 119:358–382. Publisher Copyright: © 2019 Elsevier Inc. Copyright: Copyright 2020 Elsevier B.V., All rights reserved.

Macaulay, Stewart. 1963. "Non-contractual relations in business: A preliminary study." *American Sociological Review* 28(1):55–67.

Macchiavello, Rocco and Ameet Morjaria. 2015. "The Value of Relationships: Evidence from a Supply Shock to Kenyan Rose Exports." *American Economic Review* 105(9):2911–45.

Macchiavello, Rocco and Ameet Morjaria. 2023. Relational Contracts: Recent Empirical Advancements and Open Questions. NBER Working Paper 30978.

MacLeod, W. Bentley. 2007. "Reputations, relationships, and contract enforcement." *Journal of Economic Literature* 45(3):595–628.

MacLeod, W. Bentley and James M. Malcomson. 1989. "Implicit contracts, incentive compatibility, and involuntary unemployment." *Econometrica* 57(2):447–480.

MacLeod, W. Bentley and James M. Malcomson. 1998. "Motivation and markets." *American Economic Review* pp. 388–411.

Macneil, I.R. 1978. "Contracts: Adjustment of long-term economic relations under classical and neoclassical, and relational contract law." *Northwestern University Law Review* 72:854–905.

Malcomson, James M. 2013. Relational incentive contracts. In *The Handbook of Organizational Economics*. Princeton University Press pp. 1014–1065.

Milgrom, Paul R, Douglass C North and Barry R Weingast. 1990. "The role of institutions in the revival of trade: The law merchant, private judges, and the champagne fairs." *Economics & Politics* 2(1):1–23.

Muehlheusser, Gerd, Matthias Fahn and W. Bentley MacLeod. 2023. "Editorial Preface." *Editorial Preface* 179(- 34):413 – 440.

Powell, Michael. 2019. "Productivity and credibility in industry equilibrium." *RAND Journal of Economics* 50(1):121–146.

Raith, Michael. 2022. "A Formal Theory of Employment vs. Contracting." *Working Paper* .

Ramey, Garey and Joel Watson. 2002. "Contractual intermediaries." *Journal of Law, Economics, and organization* 18(2):362–384.

Shapiro, Carl and Joseph E. Stiglitz. 1984. "Equilibrium unemployment as a worker discipline device." *American Economic Review* 74(3):433–444.

Simon, Herbert A. 1951. "A formal theory of the employment relationship." *Econometrica* 19(3):293–305.

Song, Jae, David J. Price, Fatih Guvenen, Nicholas Bloom and Till Von Wachter. 2019. "Firming up inequality." *The Quarterly Journal of Economics* 134(1):1–50.

Stigler, George J. 1951. "The division of labor is limited by the extent of the market." *Journal of Political Economy* 59(3):185–193.

Tadelis, Steven and Oliver E. Williamson. 2012. Transaction Cost Economics. In *The Handbook of Organizational Economics*, ed. Robert Gibbons and John Robberts. Princeton University Press p. 159–90.

Tirole, Jean. 1988. *The theory of industrial organization*. The MIT Press.

Troya-Martinez, Marta and Liam Wren-Lewis. 2023. "Managing relational contracts." *Journal of the European Economic Association* 21(3):941–986.

Wernerfelt, Birger. 1997. "On the nature and scope of the firm: An adjustment-cost theory." *The Journal of Business* 70(4):489–514.

Wernerfelt, Birger. 2015. "The comparative advantages of firms, markets and contracts: a unified theory." *Economica* 82(326):350–367.

Wernerfelt, Birger. 2016. *Adaptation, specialization, and the theory of the firm: Foundations of the resource-based view*. Cambridge University Press.

Williamson, Oliver E. 1985. *The economic institutions of capitalism: Firms, markets,relational contracting*. New York: Free Press.

Yang, Huanxing. 2008. "Efficiency Wages and Subjective Performance Pay." *Economic Inquiry* 46(2):179–196.

A Appendix

We omit the subscript i in the proofs. We first establish the following two lemmas.

Lemma A.1. *Suppose an optimal bilateral relational contract exists. Under this contract, if $\beta > 0$, then the following three conditions hold: 1) $w_0 = 0$, 2) both (B-IC1) and (B-IC0) bind, and 3) (P-IC1) is slack.*

Proof. Prove by contradiction. Suppose that $w_0 > 0$. Since there is an excess of producers in the frictionless matching market, $\overline{\Pi}_1 = \Pi_1$. So

$$\begin{aligned}\Pi_0 &= -w_0 + \delta\Big[\alpha_0\Pi_1 + (1-\alpha_0)(\beta\Pi_0 + (1-\beta)\overline{\Pi}_0)\Big] \\ &= -w_0 + \delta\Big[\alpha_0\overline{\Pi}_1 + (1-\alpha_0)(\beta\Pi_0 + (1-\beta)\overline{\Pi}_0)\Big].\end{aligned}$$

Also note that $\overline{\Pi}_0 = \delta\Big[\alpha_0\overline{\Pi}_1 + (1-\alpha_0)\overline{\Pi}_0\Big]$. Therefore,

$$\Pi_0 - \overline{\Pi}_0 = -w_0 + \delta(1-\alpha_0)\beta(\Pi_0 - \overline{\Pi}_0) < \delta(1-\alpha_0)\beta(\Pi_0 - \overline{\Pi}_0),$$

where the inequality comes from $w_0 > 0$. Since $\delta \in (0,1)$ and $\beta \in (0,1]$, we discuss whether $\alpha_0 = 1$. If $\alpha_0 < 1$, the equation above cannot be satisfied, which contradicts that $w_0 > 0$. On the other hand, if $\alpha_0 = 1$, $\overline{\Pi}_0 = \delta\overline{\Pi}_1$ and then $\Pi_0 = -w_0 + \delta\overline{\Pi}_1 < \overline{\Pi}_0$, which contradicts to (B-IC0) and also indicates that $w_0 = 0$. Therefore, $w_0 = 0$ and $\Pi_0 = \overline{\Pi}_0$. Thus both (B-IC1) and (B-IC0) bind.

We now show that (P-IC1) is slack. Suppose it binds, namely $U_1 = \overline{U}$. Then given $w_0 = 0$ and $\delta < 1$, plug in $U_1 = \overline{U}$ and get

$$\begin{aligned}U_0 &= \delta\Big[\alpha_0\overline{U} + (1-\alpha_0)(\beta U_0 + (1-\beta)\overline{U})\Big] \\ &< (1-\alpha_0)\beta U_0 + (1-(1-\alpha_0)\beta)\overline{U}.\end{aligned}$$

The inequality above indicates that $U_0 < \overline{U}$, which contradicts to (P-IC0). So (P-IC1) is slack. □

Lemma A.2. *For any buyer who directly contracts with producers, maximizing* Π_1 *is equivalent to minimizing* w_1.

Proof. Based on Lemma A.1, the buyer's continuation payoffs can be written as

$$\Pi_1 = \frac{1-\delta(1-\alpha_0)}{1-\delta\alpha_0\alpha_1-\delta(2-\alpha_0-\alpha_1)+\delta^2(1-\alpha_0)(1-\alpha_1)}(y-w_1),$$

$$\Pi_0 = \frac{\delta\alpha_0}{1-\delta(1-\alpha_0)}\Pi_1.$$

Since $\frac{\partial \Pi_1}{\partial w_1} < 0$, a buyer's problem is equivalent to minimize w_1 subject to producer's incentive constraints. □

Proof of Lemma 1

For simplicity, we write w_1 as w in the rest of the proof. We complete the proof by analyzing and comparing the terms in the optimal bilateral relational contracts when choosing $\beta = 0$ or $\beta > 0$.

Choice 1: $\beta = 0$. If the optimal choice of β is 0, $U_1 = w - c + \delta\Big[(1-\alpha_1)U_1 + \alpha_1\overline{U}\Big]$. The buyer optimally chooses w subject to a binding (P-IC-e), which gives

$$w_{DS} = \frac{1}{\delta(1-\alpha_1)}c + (1-\delta)\overline{U}.$$

Choice 2: $\beta \in (0,1]$. If the optimal choice of β is greater than 0, a buyer's optimization problem becomes $\min_{w,\beta} w$ subject to (P-IC-e), (P-IC0), (1), (2), $0 < \beta \leq 1$, and $w \geq 0$.

The Lagrangian is given by

$$\begin{aligned} L = w &+ \lambda_1(U_1 - (w - c + \delta((1-\alpha_1)\cdot U_1 + \alpha_1(\beta U_0 + (1-\beta)\overline{U})))) \\ &+ \lambda_2(U_0 - \delta(\alpha_0\cdot U_1 + (1-\alpha_0)\cdot(\beta U_0 + (1-\beta)\overline{U}))) \\ &+ \mu_1(w + \delta\overline{U} - U_1) + \mu_2(\overline{U} - U_0) + \mu_3(\beta - 1) + \mu_4(-w). \end{aligned}$$

The Kuhn-Tucker conditions are given by

$$\frac{\partial L}{\partial w} = 1 + \mu_1(1 - \frac{\partial U_1}{\partial w}) - \mu_2 \frac{\partial U_0}{\partial w} - \mu_4 \leq 0,$$

$$\frac{\partial L}{\partial w} w = 0,$$

$$\frac{\partial L}{\partial \beta} = -\mu_1 \frac{\partial U_1}{\partial \beta} - \mu_2 \frac{\partial U_0}{\partial \beta} - \mu_3 = 0,$$

$$\lambda_1, \lambda_2 > 0,$$

$$\mu_1, \mu_2, \mu_3, \mu_4 \geq 0,$$

$$\mu_1(w + \delta \overline{U} - U_1) = 0,$$

$$\mu_2(\overline{U} - U_0) = 0,$$

$$\mu_3(1 - \beta) = 0,$$

$$\mu_4 w = 0.$$

Meanwhile, by taking derivatives on both sides of equations (1) and (2) with respect to w, get

$$\frac{\partial U_1}{\partial w} = 1 + \delta \Big[(1 - \alpha_1) \frac{\partial U_1}{\partial w} + \alpha_1 \beta \frac{\partial U_0}{\partial w} \Big], \tag{A1}$$

$$\frac{\partial U_0}{\partial w} = \delta \Big[\alpha_0 \frac{\partial U_1}{\partial w} + (1 - \alpha_0) \beta \frac{\partial U_0}{\partial w} \Big]. \tag{A2}$$

By taking derivatives on both sides of equations (1) and (2) with respect to β, get

$$\frac{\partial U_1}{\partial \beta} = \delta \Big[(1 - \alpha_1) \frac{\partial U_1}{\partial \beta} + \alpha_1 (U_0 - \overline{U} + \beta \frac{\partial U_0}{\partial \beta}) \Big], \tag{A3}$$

$$\frac{\partial U_0}{\partial \beta} = \delta \Big[\alpha_0 \frac{\partial U_1}{\partial \beta} + (1 - \alpha_0)(U_0 - \overline{U} + \beta \frac{\partial U_0}{\partial \beta}) \Big]. \tag{A4}$$

We proceed by the following steps.

Step 1: Show that $\mu_4 = 0$ and $w > 0$. Suppose that $w = 0$. Then

$$
\begin{aligned}
U_1 &= -c + \delta\Big[(1-\alpha_1)U_1 + \alpha_1(\beta U_0 + (1-\beta)\overline{U})\Big] \\
&< (1-\alpha_1)U_1 + \alpha_1(\beta U_0 + (1-\beta)\overline{U}),
\end{aligned}
$$

where the inequality comes from that $c > 0$ and $\delta < 1$. The inequality above implies that

$$
U_1 < \beta U_0 + (1-\beta)\overline{U} < U_0,
$$

where the second inequality comes from (P-IC0). Meanwhile,

$$
\begin{aligned}
U_0 &= \delta\Big[\alpha_0 U_1 + (1-\alpha_0)(\beta U_0 + (1-\beta)\overline{U})\Big] \\
&< \alpha_0 U_1 + (1-\alpha_0)(\beta U_0 + (1-\beta)\overline{U}) \\
&< U_0,
\end{aligned}
$$

where the first inequality comes from $\delta < 1$ and the second inequality comes from (P-IC0) and $U_1 < U_0$. Since $U_0 < U_0$ can never be true, we know that $w > 0$ and thus $\mu_4 = 0$.

The implication for $w > 0$ is that

$$
\frac{\partial L}{\partial w} = 1 + \mu_1(1 - \frac{\partial U_1}{\partial w}) - \mu_2 \frac{\partial U_0}{\partial w} = 0,
$$

Step 2: Discuss the values of μ_1 and μ_2. **Case 1)** $\mu_1 = \mu_2 = 0$. In this case, $\frac{\partial L}{\partial w} = 0$ is violated, indicating that this case is not possible. In other words, at least one of two incentive compatibility constraints bind.

Case 2) $\mu_1 = 0$ and $\mu_2 > 0$. In this case, $U_1 > w + \delta\overline{U}$ and $U_0 = \overline{U}$. Solve U_1 and w based on equation (1) and get

$$
U_1 = \frac{1 - \delta(1-\alpha_0)}{\delta\alpha_0}\overline{U},
$$

$$
w = c + \frac{(1-\delta)^2 - \delta(1-\delta)(\alpha_0+\alpha_1)}{\delta\alpha_0}\overline{U}.
$$

So $\frac{\partial U_1}{\partial \beta} = \frac{\partial U_0}{\partial \beta} = 0$. Meanwhile, $\frac{\partial L}{\partial w} = 0$ and $\mu_1 = 0$ imply that $1 = \mu_2 \frac{\partial U_0}{\partial w}$. $\frac{\partial L}{\partial \beta} = 0$ implies $\mu_3 = 0$, namely $\beta < 1$.

There are two conditions that need to be satisfied for this case to be feasible and optimal. First, for feasibility, the solved w and U_1 need to satisfy $U_1 > w + \delta \overline{U}$. After plugging in U_1 and w as functions of $\overline{U}$, this requires that

$$\frac{(1-\delta)\overline{U}}{c} > \frac{\alpha_0}{1-\alpha_1}.$$

Second, since we are considering the case where choosing $\beta > 0$ is weakly better than choosing $\beta = 0$, the solved w needs to be lower than w_{DS}, which gives

$$\frac{(1-\delta)\overline{U}}{c} \leq \frac{\alpha_0}{1-\alpha_1}.$$

These two conditions contract each other, indicating that this case is not possible.

Case 3) $\mu_1 > 0$ and $\mu_2 > 0$. In this case, both incentive compatibility constraints bind, which give $U_1 = w + \delta \overline{U}$ and $U_0 = \overline{U}$. Under binding (P-IC-e) and (P-IC0), equations (1) and (2) can be satisfied only if

$$\frac{(1-\delta)\overline{U}}{c} = \frac{\alpha_0}{1-\alpha_1}.$$

In that case, the solved w also coincides with w_{DS}, and a buyer is thus indifferent among choosing any value of $\beta \in [0, 1]$.

Case 4) $\mu_1 > 0$ and $\mu_2 = 0$. In this case, $U_1 = w + \delta \overline{U}$, $U_0 > \overline{U}$, $\frac{\partial L}{\partial w} = 1 + \mu_1 (1 - \frac{\partial U_1}{\partial w}) = 0$, and $\frac{\partial L}{\partial \beta} = -(\mu_1 \frac{\partial U_1}{\partial \beta} + \mu_3) = 0$.

We discuss whether $\beta = 1$ or not. If $\beta \neq 1$, $\mu_3 = 0$, then $\frac{\partial U_1}{\partial \beta} = 0$. Given that, equation (A3) suggests that $\frac{\partial U_0}{\partial \beta} < 0$, while equation (A4) suggests that $\frac{\partial U_0}{\partial \beta} > 0$. Therefore, a contradiction exists, indicating that $\beta = 1$.

Given $\beta = 1$ and $\mu_3 > 0$, solve

$$w_{DI} = \left(\frac{1}{\delta} \frac{1 + \frac{\delta}{1-\delta}\alpha_0}{(1-\alpha_1) + \frac{\delta}{1-\delta}\alpha_0} \right) c + \left(\frac{1 + \frac{\delta}{1-\delta}\alpha_0}{(1-\alpha_1) + \frac{\delta}{1-\delta}\alpha_0} - \delta \right) \overline{U}.$$

In sum, if a buyer decides to choose $\beta > 0$, she will optimally choose $\beta = 1$ with paying w_{DI} when her demand is 1. The comparison between choice 1 and choice 2 hinge on comparing w_{DS} and w_{DI}. It turns out that $w_{DS} < w_{DI}$ if and only if

$$\frac{(1-\delta)\overline{U}}{c} > \frac{\alpha_0}{1-\alpha_1}.$$

Therefore, whenever the condition above holds, the buyer chooses $\beta = 0$ (separating from the producer when demand becomes 0) and pays w_{DS} when her demand is 1. Otherwise, she chooses $\beta = 1$ (retaining the producer when demand becomes 0) and pays w_{DI} when her demand is 1.

Proof of Lemma 2

Observe that the continuation payoffs and incentive compatibility constraints for a manager in a managed contract are similar with those for a producer in a direct contract. The only differences are that the "cost of production" for a manager is w_M and its continuation value after separating from a buyer, given by $\overline{V}$, is 0. Since

$$\frac{(1-\delta)\overline{V}}{w_M} \leq \frac{\alpha_0}{1-\alpha_1}$$

for any α_0 and α_1. Therefore, the value of p is determined by replacing c with w_M and $\overline{U}$ with 0 in w_{DI}.

Proof of Proposition 1

We prove the proposition in four steps.

Step 1: Compare direct contracting with separation with direct contracting with idleness. The following lemma comes directly from Lemma 1.

Lemma A.3. *If $\overline{U} > \overline{U}^{ei} \equiv \frac{c}{1-\delta}\alpha_0$, there exists a unique cutoff $\overline{\alpha}_1^{ei} \in (0,1)$ such that $w_{DS} < w_{DI}$ if and only if $\alpha_1 < \overline{\alpha}_1^{ei}$.*

Step 2: Compare direct contracting with separation with managed contracting.

Lemma A.4. *There exists a unique cutoff* $\overline{\alpha}_1^{eo} \in (0, 1]$ *such that* $w_{DS} < p$ *if and only if* $\alpha_1 < \overline{\alpha}_1^{eo}$.

Proof. Observe that $w_{DS} < p$ if and only if

$$\frac{1}{\delta}\frac{1}{1-\alpha_1}c + (1-\delta)\overline{U} < (\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_0}{(1-\alpha_1)+\frac{\delta}{1-\delta}\alpha_0})\frac{c}{\delta} + (\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_0}{(1-\alpha_1)+\frac{\delta}{1-\delta}\alpha_0})(1-\delta)\overline{U},$$

or

$$\frac{1}{1-\alpha_1}\frac{c}{\delta} + (1-\delta)\overline{U} - \frac{1}{\delta - \frac{\delta}{1+\frac{\delta}{1-\delta}\alpha_0}\alpha_1}(\frac{c}{\delta} + (1-\delta)\overline{U}) < 0.$$

Let the LHS to be $f(\alpha_1) = \frac{1}{1-\alpha_1}\frac{c}{\delta} + (1-\delta)\overline{U} - \frac{1}{\delta - \frac{\delta}{1+\frac{\delta}{1-\delta}\alpha_0}\alpha_1}(\frac{c}{\delta} + (1-\delta)\overline{U})$. Observe that, when $\alpha_0 = 0$, $f(\alpha_1)\big|_{\alpha_0=0} = (1-1/\delta)\frac{c}{\delta}\frac{1}{1-\alpha_1} + (1 - \frac{1}{\delta(1-\alpha_1)})(1-\delta)\overline{U} < 0$. In this case, $w_{DS} < p$ for sure, so $\overline{\alpha}_1^{eo} = 1$.

When $\alpha_0 > 0$, observe that $f(0) = (1-\frac{1}{\delta})\kappa < 0$, and $f(1)$ goes to infinity. We now show that the continuous function $f(\alpha_1)$ intersects with 0 only once. Compute

$$\frac{\partial f(\alpha_1)}{\partial \alpha_1} = \frac{\zeta(\zeta\frac{c}{\delta} - \kappa)\alpha_1^2 - 2(c-\kappa)\zeta\alpha_1 + (\delta c - \zeta\kappa)}{(1-\alpha_1)^2(\delta - \zeta\alpha_1)^2}.$$

where $\zeta = \frac{\delta}{1+\frac{\delta}{1-\delta}\alpha_0}$ and $\kappa = \frac{c}{\delta} + (1-\delta)\overline{U}$.

Observe that the numerator is a quadratic equation, where the coefficient of α_1^2 is negative, the coefficient of α_1 is positive, and the constant $\delta c - \zeta(\frac{c}{\delta} + (1-\delta)\overline{U})$ can be positive or negative. Therefore, $f(\alpha_1)$ is either strictly increasing, or is first decreasing then increasing. In either case, $f(\alpha_1)$ intersects with 0, with the intersect being $\overline{\alpha}_1^{eo} \in (0, 1)$.

In sum, there exists a unique cutoff $\overline{\alpha}_1^{eo} \in (0, 1]$ such that $w_{DS} < p$ if and only if $\alpha_1 < \overline{\alpha}^{eo}$. $\square$

Step 3: Compare direct contracting with idleness with managed contracting.

Lemma A.5. *If* $\overline{U} > \overline{U}^{io} \equiv \frac{(1-\delta(1-\alpha_0))c}{\delta(\delta(2-(1-\delta)\alpha_0)-1)}$, *there exists a unique cutoff* $\overline{\alpha}_1^{io} \in (0, 1)$ *such that* $w_{DI} < p$ *if and only if* $\alpha_1 < \overline{\alpha}_1^{io}$. *Otherwise,* $w_{DI} < p$ *for sure.*

Proof. Observe that $w_{DI} < p$ if and only if

$$(\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_0}{(1-\alpha_1)+\frac{\delta}{1-\delta}\alpha_0})c + (\frac{1+\frac{\delta}{1-\delta}\alpha_0}{(1-\alpha_1)+\frac{\delta}{1-\delta}\alpha_0} - \delta)\overline{U} < (\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_0}{(1-\alpha_1)+\frac{\delta}{1-\delta}\alpha_0})(\frac{c}{\delta} + (1-\delta)\overline{U}),$$

or

$$\frac{1+\frac{\delta}{1-\delta}\alpha_0}{1+\frac{\delta}{1-\delta}\alpha_0-\alpha_1}\left(\frac{2\delta-1}{\delta}\overline{U}-\frac{1-\delta}{\delta}\frac{c}{\delta}\right)<\delta\overline{U}.$$

Let the LHS to be $g(\alpha_1)=\frac{1+\frac{\delta}{1-\delta}\alpha_0}{1+\frac{\delta}{1-\delta}\alpha_0-\alpha_1}(\frac{2\delta-1}{\delta}\overline{U}-\frac{1-\delta}{\delta}\frac{c}{\delta})$. Observe that $g(0)=\frac{2\delta-1}{\delta}\overline{U}-\frac{1-\delta}{\delta}\frac{c}{\delta}$ and $g(1)=\frac{1+\frac{\delta}{1-\delta}\alpha_0}{\frac{\delta}{1-\delta}\alpha_0}(\frac{2\delta-1}{\delta}\overline{U}-\frac{1-\delta}{\delta}\frac{c}{\delta})$. Also observe that $g(0)-\delta\overline{U}=\frac{-(1-\delta)^2}{\delta}\overline{U}-\frac{1-\delta}{\delta}\frac{c}{\delta}<0$.

If $\overline{U}<\overline{U}^{io}$, $g(1)<\delta\overline{U}$ and thus the inequality holds for sure. Otherwise, since $g(\alpha_1)$ is strictly increasing in α_1, there exists a unique cutoff $\overline{\alpha}_1^{io}$ by the intermediate value theorem. □

Step 4: Let $\overline{U}^*=\overline{U}^{io}$ and $\alpha^*=\min\{\overline{\alpha}_1^{io},\overline{\alpha}_1^{eo}\}$.

If $\overline{U}<\overline{U}^{io}$, $w_{DI}<p$ by Lemma A.5. Otherwise, if $\overline{U}\geq\overline{U}^{io}$ there are two cases. If $w_{DS}<w_{DI}$, managed contract is optimal when $p<w_{DS}$, namely when $\alpha_1>\overline{\alpha}_1^{eo}$. If $w_{DS}>w_{DI}$ managed contract is optimal when $p<w_{DI}$, namely when $\alpha_1>\overline{\alpha}_1^{io}$.

Proof of Corollary 1

First, note that $w_D(\alpha_i)>w_M$ for all $i\in\mathcal{I}_{DS}\cup\mathcal{I}_{DI}$. Therefore, $E[w_D(\alpha_i)\mid i\in\mathcal{I}_{DS}\cup\mathcal{I}_{DI}]>E[w_M|i\in\mathcal{I}_M]$. Second, note that $w_D(\alpha_i)$ takes on different values depending on α_i, which is heterogeneous across buyers, while w_M is constant. Therefore, $\mathrm{Var}[w_D(\alpha_i)\mid i\in\mathcal{I}_{DS}\cup\mathcal{I}_{DI}]>\mathrm{Var}[w_M\mid i\in\mathcal{I}_M]=0$. Third, the separation rate is given by $\beta_i\alpha_{1i}$. Note that $\beta_i=1$ for all $i\in\mathcal{I}_{DI}$, while $\beta_i=0$ for all $i\in\mathcal{I}_{DS}\cup\mathcal{I}_M$. Fourth, the idleness for a buyer $i\in\mathcal{I}_{DI}$ is given by $(1-\pi_i)(1-\omega_i)>0$. By contrast, for $i\in\mathcal{I}_{DS}\cup\mathcal{I}_M$, idleness is zero.

Proof of Corollary 2

With the introduction of managers, the per-period payoff of buyer $i\in\mathcal{S}_0$ increases by $y-p(\alpha_i)$. The per-period payoff of buyer $i\in\mathcal{S}_I$ increases by $w_{DI}(\alpha_i)-p(\alpha_i)$. The per-period payoff of buyer $i\in\mathcal{S}_S$ increases by $w_{DS}(\alpha_i)-p(\alpha_i)$. The per-period payoff of the remaining buyers are unchanged.

The average producer pay per unit effort falls, since $w_D(\alpha_i)>w_M$ for all $i\in\mathcal{S}_I\cup\mathcal{S}_S$. Producer separation rates and idleness also fall, since buyers switch from direct contracts with either idleness and separation to managed contracts in which buyers are never idle and never separate.

The measure of buyers who receive services increases by $|\mathcal{S}_0|$. The total measure of services provided in the economy increases by $\int_{\mathcal{S}_0} \pi_i dF$. However, for $i \in \mathcal{S}_I$, the steady-state measure of producers matched to these buyers fall, since idleness falls. The total measure of matched producer ($n_M + n_B$) therefore increases if and only if $\int_{\mathcal{S}_0} \pi_i dF > \int_{\mathcal{S}_I} (1-\pi_i) dF$.

Proof of Proposition 4

Based on Lemma 1 and 2, the payments by buyer i under direct and managed contracts are

$$w_D(\alpha_i) = \frac{1}{\delta}\frac{1}{1-\alpha_i}c + (1-\delta)\overline{U},$$

$$w_G = w_M = \frac{c}{\delta} + (1-\delta)\overline{U},$$

$$p(\alpha_i) = \frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_i}{1+\frac{\delta}{1-\delta}\alpha_i - \alpha_i}\left[\frac{c}{\delta} + (1-\delta)\overline{U}\right],$$

where $w_D(\alpha_i)$ is the pay for a specialist, w_G is the pay for a generalist, w_M is the pay from a manager to a managed producer, and $p(\alpha_i)$ is the service fee for managers.

The buyer's post-matching continuation payoffs when she has demand when choosing to directly contract with specialists, to directly contract with a generalist, or to outsource are thus

$$\Pi_B(\alpha_i) = \frac{y+\Delta_i}{1-\delta} - \left[\overline{U} + \frac{c}{(1-\alpha_i)\delta(1-\delta)}\right],$$

$$\Pi_G = \frac{y}{1-\delta} - \left[\overline{U} + \frac{c}{\delta(1-\delta)}\right].$$

$$\Pi_M(\alpha_i) = \frac{y+\Delta_i}{1-\delta} - \left[\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_i}{1+\frac{\delta}{1-\delta}\alpha_i - \alpha_i}\left(\overline{U} + \frac{c}{\delta(1-\delta)}\right)\right],$$

respectively.

Three observations follow. First, a buyer prefers to directly contract with a specialist than directly contract with a generalist if and only if $\Pi_B(\alpha_i) \geq \Pi_G$, or,

$$\Delta_i \geq \frac{\alpha_i}{1-\alpha_i}\frac{c}{\delta}.$$

Second, a buyer prefers to directly contract with specialists rather than contract with a manager if and only if

$$\left[\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_i}{1+\frac{\delta}{1-\delta}\alpha_i-\alpha_i}-1\right]\left[\overline{U}+\frac{c}{\delta(1-\delta)}\right] \geq \frac{\alpha_i}{1-\alpha_i}\frac{c}{\delta(1-\delta)}.$$

By Lemma A.3, there exists a unique cutoff α^{ei} such that the buyer prefers to contract with a manager if and only if $\alpha_i > \alpha^{ei}$. Third, a buyer prefers a managed contract over direct contract with a generalist if and only if

$$\Delta_i \geq \overline{\Delta}(\alpha_i) \equiv \left[\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_i}{1+\frac{\delta}{1-\delta}\alpha_i-\alpha_i}-1\right]\left[\overline{U}+\frac{c}{\delta(1-\delta)}\right].$$

Therefore, the buyer choose to contract with a manager if and only if α_i and Δ_i are both large enough.

Proof of Corollary 3

Let $\mathcal{S}_G$ denote buyers who switch from direct contracts with generalists to managed contracts with specialists when managers become available. Let $\mathcal{S}_B$ be the set of buyers who switch from direct contracts with specialists to managed contracts with specialists. Given the assumption on the distribution of (Δ_i, α_i), $|\mathcal{S}_G|, |\mathcal{S}_B| > 0$. The contractual choice of the remaining buyers are unchanged. Therefore, the measure of specialists unambiguously increase with managerial coordination.

Note that the Bellman equation for unmatched producers can be rewritten as

$$\overline{U} = \frac{v}{u}\frac{E[v_i U_{1i}]}{v} + \left(1-\frac{v}{u}\right)\delta\overline{U}.$$

With the introduction of managers, more producers are contracted as specialists under managed contracts without separation, so v and $E[v_i U_{1i}]$ both fall. This implies that $\frac{v}{u}$ increases. It follows that the measure of unmatched producers $n_N = u - v$ falls.

Proof of Proposition 5

In a multi-task economy with reputable managers, the service fee is given by

$$p(\alpha_i) = \left(\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_i}{(1-\alpha_i)+\frac{\delta}{1-\delta}\alpha_i}\right) w_M + \left(\frac{1+\frac{\delta}{1-\delta}\alpha_i}{(1-\alpha_i)+\frac{\delta}{1-\delta}\alpha_i} - \delta\right)(-\gamma\tilde{V})$$

This implies that

$$\Pi_M(\alpha_i) = \frac{y+\Delta_i}{1-\delta} - \Big[\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_i}{1+\frac{\delta}{1-\delta}\alpha_i - \alpha_i}(\overline{U} + \frac{c}{\delta(1-\delta)})\Big] + \left(\frac{1+\frac{\delta}{1-\delta}\alpha_i}{(1-\alpha_i)+\frac{\delta}{1-\delta}\alpha_i} - \delta\right)\frac{\gamma\tilde{V}}{1-\delta},$$

As γ increases, $\Pi_M(\alpha_i)$ rises, so buyers switch to managed contracts with specialists from direct contracts with both generalists and specialists. Therefore, by the same logic as Corollary 3, the measure of specialists increases, while the measure of unmatched producers decreases.

B Online Appendix

In this section, we show that, in our model with limited liability, it is without loss of generality to focus on analyzing stationary relational contracts without bonus payments given $\overline{U} > 0$. To do so, we first show that, when the limited liability constraints are not imposed, it is without loss to focus on stationary relational contracts without bonus payments, following the idea in Sections A.1 and A.2 of Board and Meyer-ter Vehn (2015). We then show that assuming limited liability is without loss.

Consider a model without limited liability where the buyer can pay the produce a bonus after observing the effort. In that model, suppose $\beta_t > 0$, $\forall t$, the continuation payoffs of producers and buyers (omitting subscription i) are

$$U_{1,t} = w_{1,t} - c(e_t) + b_{1,t} + \delta\Big[(1-\alpha_1)U_{1,t+1} + \alpha_1(\beta_t U_{0,t+1} + (1-\beta_t)\overline{U})\Big],$$

$$U_{0,t} = w_{0,t} + b_{0,t} + \delta\Big[\alpha_0 U_{1,t+1} + (1-\alpha_0)(\beta_t U_{0,t+1} + (1-\beta_t)\overline{U})\Big],$$

$$\Pi_{1,t} = y(e_t) - w_{1,t} - b_{1,t} + \delta\Big[(1-\alpha_1)\Pi_{1,t+1} + \alpha_1(\beta_t \Pi_{0,t+1} + (1-\beta_t)\overline{\Pi}_0)\Big],$$

$$\Pi_{0,t} = -w_{0,t} - b_{0,t} + \delta\Big[\alpha_0 \Pi_{1,t+1} + (1-\alpha_0)(\beta_t \Pi_{0,t+1} + (1-\beta_t)\overline{\Pi}_0)\Big],$$

respectively. Note that if $\beta_\tau = 0$ for some τ, there is no $U_{0,t}$ and $\Pi_{0,t}$ for $t > \tau$.

Suppose there exists a relational contract $(e_t, w_{1,t}, b_{1,t}, w_{0,t}, b_{0,t}, \beta_t)_t$ that satisfies the following constraints:

$$U_{1,t} \geq w_{1,t} + \delta\overline{U},$$

$$U_{1,t} \geq \overline{U},$$

$$U_{0,t} \geq \overline{U},$$

$$\Pi_{1,t} \geq \delta(\alpha_1\overline{\Pi}_0 + (1-\alpha_1)\overline{\Pi}_1),$$

$$\Pi_{1,t} \geq \overline{\Pi}_1,$$

$$\Pi_{0,t} \geq \overline{\Pi}_0,$$

$$-b_{1,t} + \delta\Big[(1-\alpha_1)\Pi_{1,t+1} + \alpha_1(\beta_t \Pi_{0,t+1} + (1-\beta_t)\overline{\Pi}_0)\Big] \geq \delta(\alpha_1\overline{\Pi}_0 + (1-\alpha_1)\overline{\Pi}_1),$$

$$-b_{0,t} + \delta\Big[\alpha_0 \Pi_{1,t+1} + (1-\alpha_0)(\beta_t \Pi_{0,t+1} + (1-\beta_t)\overline{\Pi}_0)\Big] \geq \delta(\alpha_0\overline{\Pi}_1 + (1-\alpha_0)\overline{\Pi}_1).$$

Compared to the constraints listed in Section 3, allowing bonus payments does not introduce any additional constraints for the producer because the original (P-IC-e) has captured the producer's incentive to exert efforts. The additional constraints for the buyer requires that the continuation values of keeping the producer whenever the producer has demand is higher than the bonus paid in either situation. Note again that if $\beta_\tau = 0$ for some τ, the constraints on $\Pi_{0,t}$ does not exist for $t > \tau$.

B.1 Irrelevance of bonus payments

Based on the relational contract with bonus payments $(e_t, w_{1,t}, b_{1,t}, w_{0,t}, b_{0,t}, \beta_t)_t$, we can construct a new relational contract without bonus payments that generates the same continuation payoffs for the buyer, $(\hat{w}_{1,t}, \hat{w}_{0,t}, \hat{\beta}_t)_t$, by shifting the bonus payments in the last period to the next period with adjusting to discounting and potential separation.

Specifically, let $\hat{w}_{1,t} = w_{1,t} + \frac{(1-\alpha_0)\frac{b_1}{\delta} - \alpha_1 \frac{b_0}{\delta}}{1-\alpha_0-\alpha_1}$, $\hat{w}_{0,t} = w_{0,t} + \frac{(1-\alpha_1)\frac{b_0}{\delta} - \alpha_0 \frac{b_1}{\delta}}{1-\alpha_0-\alpha_1}$, and $\hat{\beta}_t = \beta_t$, $\forall t$. By doing so, all the incentive constraints are satisfied since the following conditions are satisfied:

$$-b_{1,t}+\delta\Big[(1-\alpha_1)\Pi_{1,t+1}+\alpha_1(\beta_t\Pi_{0,t+1}+(1-\beta_t)\overline{\Pi}_0)\Big] = \delta\Big[(1-\alpha_1)\hat{\Pi}_{1,t+1}+\alpha_1(\hat{\beta}_t\hat{\Pi}_{0,t+1}+(1-\hat{\beta}_t)\overline{\Pi}_0)\Big]$$

$$-b_{0,t}+\delta\Big[\alpha_0\Pi_{1,t+1}+(1-\alpha_0)(\beta_t\Pi_{0,t+1}+(1-\beta_t)\overline{\Pi}_0)\Big] = \delta\Big[\alpha_0\hat{\Pi}_{1,t+1}+(1-\alpha_0)(\hat{\beta}_t\hat{\Pi}_{0,t+1}+(1-\hat{\beta}_t)\overline{\Pi}_0)\Big].$$

Under this relational contract without bonus payments, the buyer's continuation payoffs stay the same as the original relational contract with bonus payments. It is therefore without loss to focus on relational contracts without bonus payments.

B.2 No loss from assuming stationarity

To show that it is without loss of generality to focus on stationary relational contracts, we now construct a stationary relational contract $(e^*, w_1^*, w_0^*, \beta^*)$ based on any potentially non-stationary relational contract $(e_t, w_{1,t}, w_{0,t}, \beta_t)_t$ that generates a weakly higher continuation payoff for the buyer when starting a relationship, i.e., $\overline{\Pi}_1^* \geq \overline{\Pi}_1$.

Under $(e_t, w_{1,t}, w_{0,t}, \beta_t)_t$, the producer's and the buyer's continuation payoffs in a match are

$$U_{1,t} = w_{1,t} - c(e_t) + \delta\Big[(1-\alpha_1)U_{1,t+1} + \alpha_1(\beta_t U_{0,t+1} + (1-\beta_t)\overline{U})\Big],$$

$$U_{0,t} = w_{0,t} + \delta\Big[\alpha_0 U_{1,t+1} + (1-\alpha_0)(\beta_t U_{0,t+1} + (1-\beta_t)\overline{U})\Big],$$

$$\Pi_{1,t} = y(e_t) - w_{1,t} + \delta\Big[(1-\alpha_1)\Pi_{1,t+1} + \alpha_1(\beta_t\Pi_{0,t+1} + (1-\beta_t)\overline{\Pi}_0)\Big],$$

$$\Pi_{0,t} = -w_{0,t} + \delta\Big[\alpha_0\Pi_{1,t+1} + (1-\alpha_0)(\beta_t\Pi_{0,t+1} + (1-\beta_t)\overline{\Pi}_0)\Big].$$

The incentive constraints for the producer and the buyer are:

$$U_{1,t} \geq w_{1,t} + \delta\overline{U},$$

$$U_{1,t} \geq \overline{U},$$

$$U_{0,t} \geq \overline{U},$$

$$\Pi_{1,t} \geq \delta(\alpha_1\overline{\Pi}_0 + (1-\alpha_1)\overline{\Pi}_1),$$

$$\Pi_{1,t} \geq \overline{\Pi}_1,$$

$$\Pi_{0,t} \geq \overline{\Pi}_0.$$

Now we construct $(e^*, w_1^*, w_0^*, \beta^*)$. First, suppose $e_{t'} = 1$ in some period t' and let $e^* = e_{t'} = 1$ (t' exists since otherwise the original relational contract will be a stationary in the effort with effort being 0 in all periods). Second, let $U_1^* = U_{1,t'}$ and $U_0^* = U_{0,t'}$. Finally, let $\beta^* = \beta_{t'}$ and set the payments w_1^* and w_0^* such that they satisfy:

$$U_1^* = w_1^* - c + \delta\Big[(1-\alpha_1)U_1^* + \alpha_1(\beta^* U_0^* + (1-\beta^*)\overline{U})\Big],$$

$$U_0^* = w_0^* + \delta\Big[\alpha_0 U_1^* + (1-\alpha_0)(\beta^* U_0^* + (1-\beta^*)\overline{U})\Big].$$

The buyer's continuation payoffs are thus

$$\Pi_1^* = y - w_1^* + \delta\Big[(1-\alpha_1)\Pi_1^* + \alpha_1(\beta^*\Pi_0^* + (1-\beta^*)\overline{\Pi}_0)\Big],$$

$$\Pi_0^* = -w_0^* + \delta\Big[\alpha_0\Pi_1^* + (1-\alpha_0)(\beta^*\Pi_0^* + (1-\beta^*)\overline{\Pi}_0)\Big].$$

Under this new stationary relational contract, the producer's incentive constraints are satisfied. Indeed, the first incentive constraint requires

$$\delta\Big[(1-\alpha_1)(U_1^* - \overline{U}) + \alpha_1\beta^*(U_0^* - \overline{U})\Big] \geq c,$$

as $\delta\Big[(1-\alpha_1)(U_{1,t'}-\overline{U})+\alpha_1\beta_{t'}(U_{0,t'}-\overline{U})\Big] \geq c(e_{t'})$ at t'. Meanwhile, we know

$$U_1^* = U_{1,t'} \geq \overline{U},$$

$$U_0^* = U_{0,t'} \geq \overline{U},$$

The buyer's incentive constraints are satisfied since the relational contract is stationary, which implies $\Pi_1^* = \overline{\Pi}_1$ and $\Pi_0^* = \overline{\Pi}_0$ in this frictionless market with excess supply of producers.

To show $\overline{\Pi}_1^* \geq \overline{\Pi}_1$, given that $\overline{\Pi}_1^* = \Pi_1^*$ and $\Pi_{1,t'} \geq \overline{\Pi}_1$, we need only to show $\Pi_1^* \geq \Pi_{1,t'}$. We do so by comparing the joint surpluses $\Pi_1^* + U_1^*$ and $\Pi_{1,t'} + U_{1,t'}$. It is obvious that $\Pi_1^* + U_1^* \geq \Pi_{1,t'} + U_{1,t'}$, since under the stationary relational contract, effort is motivated and thus the flow payoffs are maximized in each period when there is demand. Given that $U_1^* = U_{1,t'}$ by construction, we have $\Pi_1^* \geq \Pi_{1,t'}$.

B.3 No loss from assuming limited liability

We now show that imposing limited liability is without loss. The idea is that, if the limited liability constraints are violated under an equilibrium relational contract (i.e., if the pay is below zero when there is or isn't demand), we can show that either 1) there exists another equilibrium relational contract where the limited liability constraints are not violated, or 2) the original relational contract cannot be sustained in equilibrium.

Under a stationary relational contract without bonus payments (w_1, w_0, β) (omitting effort as it is always 1), the continuation payoffs are

$$U_1 = w_1 - c + \delta\Big[(1-\alpha_1)U_1 + \alpha_1(\beta U_0 + (1-\beta)\overline{U})\Big],$$

$$U_0 = w_0 + \delta\Big[\alpha_0 U_1 + (1-\alpha_0)(\beta U_0 + (1-\beta)\overline{U})\Big],$$

$$\Pi_1 = y - w_1 + \delta\Big[(1-\alpha_1)\Pi_1 + \alpha_1(\beta\Pi_0 + (1-\beta)\overline{\Pi}_0)\Big],$$

$$\Pi_0 = -w_0 + \delta\Big[\alpha_0\Pi_1 + (1-\alpha_0)(\beta\Pi_0 + (1-\beta)\overline{\Pi}_0)\Big].$$

The incentive constraints for the producer and the buyer are

$$U_1 \geq w_1 + \delta \overline{U},$$

$$U_1 \geq \overline{U},$$

$$U_0 \geq \overline{U},$$

$$\Pi_1 \geq \delta(\alpha_1 \overline{\Pi}_0 + (1-\alpha_1)\overline{\Pi}_1),$$

$$\Pi_1 \geq \overline{\Pi}_1,$$

$$\Pi_0 \geq \overline{\Pi}_0.$$

The limited liability constraints are

$$w_1 \geq 0,$$

$$w_0 \geq 0.$$

Note that we can solve

$$\Delta U_1 = \frac{w_1 - c - (1-\delta)\overline{U}}{1-\delta(1-\alpha_1)} + \frac{\delta \alpha_1 \beta}{1-\delta(1-\alpha_1)} \Delta U_0,$$

$$\Delta U_0 = \frac{\delta \alpha_0 (w_1 - c - (1-\delta)\overline{U}) + (1-\delta(1-\alpha_1))(w_0 - (1-\delta)\overline{U})}{(1-\delta(1-\alpha_1))(1-\delta(1-\alpha_0)\beta) - \delta^2 \alpha_0 \alpha_1 \beta}.$$

where $\Delta U_1 = U_1 - \overline{U}$ and $\Delta U_0 = U_0 - \overline{U}$. Further note that, from Lemma A.1, $w_0 > 0$ cannot be sustained in equilibrium. Therefore, there are three cases where the limited liability constraints can be violated.

The first case is that both limited liability constraints are violated, i.e., $w_1 < 0$ and $w_0 < 0$. In this case, both ΔU_1 and ΔU_0 are strictly negative. The producer's incentive constraints are thus violated. It is thus without loss not to consider such a relational contract as it cannot be sustained in equilibrium.

The second case is that only the limited liability constraint on w_1 is violated, i.e., $w_1 < 0$ and $w_0 = 0$. In this case, both ΔU_1 and ΔU_0 are still strictly negative, indicating that the producer's incentive constraints are violated. Again, it is thus without loss not to consider such a relational contract as it cannot be sustained in equilibrium.

The third case is that only the limited liability constraint on w_0 is violated, i.e., $w_1 \geq 0$ and $w_0 < 0$. In this case, we increase w_0 and decrease w_1 such that w_0 can equal to 0. Specifically, based on (w_1, w_0, β), we construct a new relational contract (w_1^*, w_0^*, β^*) by setting $w_0^* = 0$ while keeping $U_1^* = U_1$. To do so we set $w_1^* = w_1 + \frac{\delta\alpha_1\beta}{1-\delta(1-\alpha_0)\beta}w_0$. By construction, $U_0^* = U_0 - \frac{w_0}{1-\delta(1-\alpha_0)\beta} > U_0$ given $w_0 < 0$. Under this new relational contract, the producer's incentive constraints are satisfied since his continuation payoffs weakly increase, and the buyer's incentive constraints are satisfied since Π_1 stays the same and $\Pi_0^* \geq \overline{\Pi}_0^*$. The latter comes from the fact that $\Pi_0^* - \overline{\Pi}_0^* = \frac{-w_0^*}{1-\delta(1-\alpha_0)\beta} = 0$ given $\Pi_1^* = \Pi_1 = \overline{\Pi}_1 = \overline{\Pi}_1^*$ and $w_0^* = 0$. Further, the buyer is indifferent between the original and the newly constructed relational contracts, as $\Pi_1^* = \Pi_1$. The last step is to check whether $w_1^* \geq 0$. If not, we have $w_1^* < 0$ and $w_0^* = 0$, which is the same as the second case above and thus cannot be sustained in equilibrium. Therefore, should the original relational contract be sustained in equilibrium, $w_1^* \geq 0$, indicating both the limited liability constraints are satisfied.

In sum, imposing limited liability constraints is without loss.